UNIVERSITI TEKNOLOGI MALAYSIA

# SECR1033-02

# COMPUTER ORGANIZATION AND ARCHITECTURE

## GROUP PROJECT

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Video link:

https://drive.google.com/file/d/1fWuZLTf0YSSS-orIhVKSZ7V_xuAAN4g7/view?usp=sharing

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# 1.0 Introduction

The evaluation of a computer with reference to its central processing unit (CPU) with predefined tests and programs is called benchmarking. These tests should be designed in a way to measure CPU performance in at least four aspects, including its throughput rate, its multiprocessing ability, and its performance on different types of loads. Benchmarking also enables the users to evaluate different CPUs, so that they can determine how better a particular CPU will perform in certain tasks.

Hence, benchmarking can be grouped into synthetic benchmark, real world benchmark, and application benchmark. First, synthetic benchmarks copy certain procedures like arithmetic calculations, data compression or rendering and expose the CPU, real performance. Synthetic benchmarks are applied in a situation when a consumer wants to compare CPUs as soon as possible. Real-world benchmarking shares with the system real-life results of a system's performance, for instance, assessing the capacity of a CPU in online surfing or video editing. Users get a better idea of how a CPU shall perform out of more likely activities. The application benchmarks are tied to the basic amount of a single application's CPU usage. This one gives the user accurate information on data productivity for a specific application.

This project entails CPU and hardware efficiency tests to be conducted. We need to concentrate on two sections for this project which for Part 1: It is for this reason that we will conduct the benchmarking of the CPU by using some of the tools on specific types of laptops and then make some findings. Part 2 necessitates the design of an assembly code (ASM code programmes) to compute and measure the time taken to execute the several loops of the provided equation. Some of the tools have been provided for the students while the benchmark results can only be accessed through downloading the programmes. Students were also informed that they should also watch certain videos in appendix A of the project to have a better understanding of how CPU benchmarking is done.

**Aim**

The rationale for this project is to perform benchmarking of several types of laptops across different generations using benchmarking software for the CPU. We will also get the results of the CPU benchmarks by testing assembly code, written and getting the time it will take to complete for each loop as shown in the equation.

**Objectives**

1. To compare and analyze the differences in the CPU performance of laptops, several CPU benchmarking software must be used.
2. To evaluate the different benchmarking results from different software pertaining to the different CPU in each laptop.
3. To finally determine the order of laptops depending on the performance of their Central Processing Units.

**Scopes (computer specification and benchmarking tools features)**
We will use different types of laptops to be benchmarked.

| Specification | Computer Type | | | |
|---|---|---|---|---|
| | Acer Aspire M5-481 PTG | VivoBook ASUSLaptop X515JA A516JA | VivoBook ASUSLaptop X415MA A416MA | Dell Inspiron 16 5260 0D03J8 |
| **Processor Type** | Intel Core i5 3337U | Intel Core i3 1005G1 | Intel Celeron N4020 CPU 1.10GHz | Intel Core i7 1260P |
| **RAM Size (GB)** | 9.8 | 12 | 8 | 16 |
| **GPU Type** | NVIDIA GeForce GT640M LE | Intel(R) UHD Graphics | Intel(R) UHD Graphics 600 | Intel(R) Iris(R) Xe Graphics |
| **Motherboard** | MA40_HX | X515JA | X415MA | 0D03J8 |
| **Bus Specs.** | PCI_Express 2.0(5.0 GT/s) | PCI-Express 3.0 (8.0 GT/s) | PCI-Express 2.0 (5.0 GT/s) | - |

In this project, we utilize CPU-Z, Geekbench, Monitor Task Manager, and MiniTool Partition Wizard to benchmark each of the laptops.

1. CPU-Z is a commonly used software to identify and monitor the application for Microsoft Windows. It can detect some components of the system such as central processing unit (CPU), RAM, motherboard, chipset and many more. It provides specific details about hardware components, which are useful for users who need to troubleshoot hardware problems, monitor system specifications, or improve computer performance. Here are some of the features that we will be focusing when troubleshoot the CPU-Z benchmarking:
   - Processor name and number, codename, process, package, and cache level.
   - Motherboard and chipset.
   - Memory type, size, timings, serial presence detect (SPD).
   - Real-time analysis of each core's intrinsic frequency and memory frequency.

2. Geekbench measures the performance of processors and devices. It helps the consumers to evaluate the various systems depending on different factors. There are two types of benchmarks for every processor – the single-core and multi-core that enable the users to distinguish between the performances of different CPUs. Some of the features for GeekBench benchmarking tools are:
   - Single-core and multi-core performance
   - Cross-platform benchmarking
   - The efficacy of GPU utilization and independent and comparative API performance.

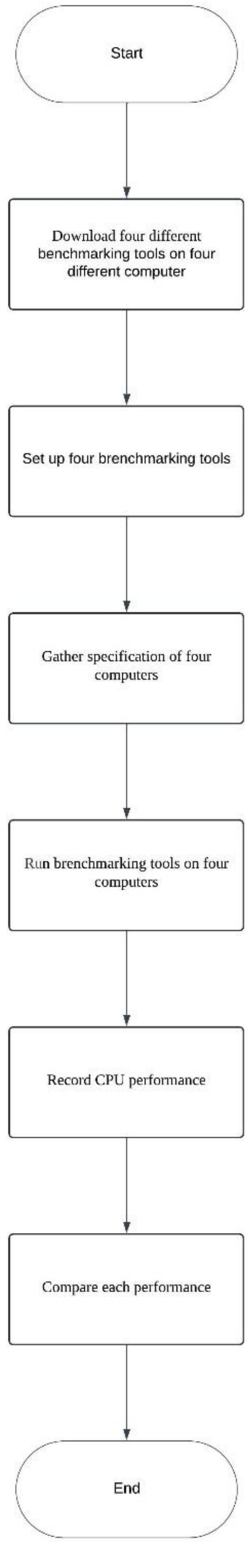
3. Monitor task manager is a fundamental software for Windows users by providing extensive tools for tracking and handling system performance, services, processes, and launching programmes. Its real-time data and extensive information make it ideal for troubleshooting, optimizing, and managing the general function of a Windows machine. Some of the features for Monitor Task Manager are listed below:
   - Usage of CPU, memory, disk, network and GPU.
   - Number of ongoing background processes.
   - Startup management.
   - App history.
   - Logged-in users and session management.

4. MiniTool Partition Wizard is a software that offers users a full set of capabilities for controlling, enhancing, and sustaining their hard disk drives (HDD) or solid-state drives (SDD) storage effectively. Some features that provided in MiniTool Partition Wizard are:
   - Partition management.
   - Disk management.
   - Data recovery.
   - Disk and partition backup.
   - Disk benchmarking.
   - File system conversion.
   - Dynamic disk management.

## 2.0

To investigate the performance of three CPUs and make an analysis to the performance data and make a comparison to the three CPUs.

# 2.0 Flowchart of Project



# 3.0 Benchmarking Results

### 3.1 Acer Aspire M5-481 PTG

| Tool | Result Details | |
|---|---|---|
| **Model** | Acer Aspire M5-481 PTG  | |
| **CPU-Z** | Motherboard | MA40_HX |
| | CPU | Intel Core i5 3337U |
| | Clock speed | 798.1 MHz |
| | Cache | L1 Cache: 128KB<br>L2 Cache: 512KB<br>L3 Cache: 3MB |
| | Memory | 10 GBytes |
| **Geekbench** | **Benchmark Score** | |
| | Single-Core | 356 |
| | Multi-Core | 717 |
| | **Performance Result** | |
| | File compression: | |
| | Single-Core | 324(46.5 MB/sec) |
| | Multi-Core | 773(111.0 MB/sec) |
| | Navigation: | |
| | Single-Core | 819 (4.93 routes/sec) |
| | Multi-Core | 1730 (10.4 routes/sec) |
| | HTML 5 Browser | |
| | Single-Core | 680 (13.9 pages/sec) |
| | Multi-Core | 1169 (23.9 pages/sec) |
| **Task manager (Low Load)** | **Processes** | |
| | Number of Active Processes | 78 |
| | **CPU Performance** | |
| | CPU Utilization (%) | 30 |
| | CPU Speed (GHz) | 2.17 |
| | No. Threads | 2241 |
| | No. Handles | 81144 |
| | **Memory Performance** | |
| | Total Memory (GB) | 9.8 |
| | Used Memory (GB) | 3.7 |
| | Memory Usage (%) | 39 |
| | In Use (Compressed) (GB) | 0.341 |

| | | |
|---|---|---|
| | Committed Memory (GB) | 4.2/11.3 |
| | Cache Memory (GB) | 5.7 |
| **Task manager (Middle Load)** | **Processes** | |
| | Number of Active Processes | 126 |
| | **CPU Performance** | |
| | CPU Utilization (%) | 54 |
| | CPU Speed (GHz) | 2.48 |
| | No. Threads | 2659 |
| | No. Handles | 102358 |
| | **Memory Performance** | |
| | Total Memory (GB) | 9.8 |
| | Used Memory (GB) | 5.9 |
| | Memory Usage (%) | 60 |
| | In Use (Compressed) (GB) | 0.56 |
| | Committed Memory (GB) | 7.9/11.3 |
| | Cache Memory (GB) | 3.5 |
| **Task manager (High Load)** | **Processes** | |
| | Number of Active Processes | 138 |
| | **CPU Performance** | |
| | CPU Utilization (%) | 100 |
| | CPU Speed (GHz) | 2.28 |
| | No. Threads | 3438 |
| | No. Handles | 120358 |
| | **Memory Performance** | |
| | Total Memory (GB) | 9.8 |
| | Used Memory (GB) | 7.3 |
| | Memory Usage (%) | 76 |
| | In Use (Compressed) (GB) | 0.223 |
| | Committed Memory (GB) | 9.1/11.3 |
| | Cache Memory (GB) | 2.4 |
| **MiniTool Partition Wizard （32KB）** | Physical Disk | KINGSTON SA400S37480 ATA |
| | **Stress Test** | |
| | Sequential Reading (MB/s) | 182.577 |
| | Sequential Writing (MB/s) | 451.567 |
| | Random Reading (MB/s) | 351.743 |
| | Random Writing (MB/s) | 409.714 |

### 3.2 VivoBook ASUSLaptop X515JA A516JA

| Tool | Result Details |
|---|---|

| | | |
|---|---|---|
| **Model** | <br>VivoBook ASUSLaptop X515JA A516JA | |
| **CPU-Z** | Motherboard | X515JA |
| | CPU | Intel Core i3 1005G1 |
| | Clock speed | 997.56 MHz |
| | Cache | L1 Cache: 160 KB<br>L2 Cache: 1024 KB<br>L3 Cache: 4 MB |
| | Memory | 12 GB DDR4 |
| **Geekbench** | **Benchmark Score** | |
| | Single-Core | 1244 |
| | Multi-Core | 2319 |
| | **Performance Result** | |
| | File Compression: | |
| | Single-Core | 1053 (151.2MB/sec) |
| | Multi-Core | 1275 (183.1 MB/sec) |
| | **Navigation:** | |
| | Single-Core | 1135 (6.84 routes/sec) |
| | Multi-Core | 2610 (15.7 routes/sec) |
| | **HTML5 Browser** | |
| | Single-Core | 1313 (26.9 pages/sec) |
| | Multi-Core | 2620 (53.6 pages/sec) |
| **Task Manager (Low Load)** | **Processes** | |
| | Member of Active Processes | 87 |
| | **CPU Performance** | |
| | CPU Utilization (%) | 49% |
| | CPU Speed (GHz) | 1.41 GHz |
| | No. Threads | 3115 |
| | No. Handles | 114410 |
| | **Memory Performance** | |
| | Total Memory (GB) | 11.7 GB |
| | Used Memory (GB) | 7.3 GB |
| | Memory Usage (%) | 60% |
| | In Use (Compressed) (GB) | 0 |
| | Committed Memory (GB) | 7.9/21.1 GB |

| | | |
|---|---|---|
| | Cache Memory (GB) | 4.1 GB |
| **Task Manager (Middle Load)** | **Processes** | |
| | Member of Active Processes | 84 |
| | **CPU Performance** | |
| | CPU Utilization (%) | 51% |
| | CPU Speed (GHz) | 1.38 GHz |
| | No. Threads | 2626 |
| | No. Handles | 106695 |
| | **Memory Performance** | |
| | Total Memory (GB) | 11.7 GB |
| | Used Memory (GB) | 6.8 GB |
| | Memory Usage (%) | 63% |
| | In Use (Compressed) (GB) | 0 |
| | Committed Memory (GB) | 7.1/21.1 GB |
| | Cache Memory (GB) | 4.8 GB |
| **Task Manager (High Load)** | **Processes** | |
| | Member of Active Processes | 87 |
| | **CPU Performance** | |
| | CPU Utilization (%) | 100% |
| | CPU Speed (GHz) | 1.67 GHz |
| | No. Threads | 3023 |
| | No. Handles | 105977 |
| | **Memory Performance** | |
| | Total Memory (GB) | 11.7 GB |
| | Used Memory (GB) | 7.2 GB |
| | Memory Usage (%) | 60% |
| | In Use (Compressed) (GB) | 0 |
| | Committed Memory (GB) | 7.0/21.1 GB |
| | Cache Memory (GB) | 2.8 |
| **MiniTool Partition Wizard (32 KB)** | Physical Disk | SAMSUNG MZVLQ256HAJD-00000 |
| | **Stress Test** | |
| | Sequential Reading (MB/s) | 455.023 |
| | Sequential Writing (MB/s) | 100.017 |
| | Random Reading (MB/s) | 259.783 |
| | Random Writing (MB/s) | 108.901 |

### 3.3 VivoBook ASUSLaptop X415MA A416MA

| Tool | Result Details | |
|---|---|---|
| **Model** | VivoBook ASUSLaptop X415MA A416MA  | |
| **CPU-Z** | Motherboard | X415MA |
| | CPU | Intel Celeron N4000 |
| | Clock Speed | 795.64 MHz |
| | Cache | L1 Cache: 48KB<br>L2 Cache: 64KB<br>L3 Cache: 4MB |
| | Memory | 8GB DDR4 |
| **Geekbench** | **Benchmark Score** | |
| | Single-Core | 329 |
| | Multi-Core | 548 |
| | **Performance Result** | |
| | File Compression: | |
| | Single-Core | 458 (65.8MB/sec) |
| | Multi-Core | 496 (71.2 MB/sec) |
| | **Navigation:** | |
| | Single-Core | 616 (3.71 routes/sec) |
| | Multi-Core | 1094 (6.59 routes/sec) |
| | **HTML5 Browser** | |
| | Single-Core | 560 (11.5 pages/sec) |
| | Multi-Core | 964 (19.7 pages/sec) |
| **Task Manager (Low Load)** | **Processes** | |
| | Member of Active Processes | 81 |
| | **CPU Performance** | |
| | CPU Utilization (%) | 38% |
| | CPU Speed (GHz) | 1.31 GHz |
| | No. Threads | 2479 |
| | No. Handles | 110868 |
| | **Memory Performance** | |

| | | |
|---|---|---|
| | Total Memory (GB) | 8.0 GB |
| | Used Memory (GB) | 7.8 GB |
| | Memory Usage (%) | 63% |
| | In Use (Compressed) (GB) | 4.9 GB (307 MB) |
| | Committed Memory (GB) | 8.3/12.0 GB |
| | Cache Memory (GB) | 2.3 GB |
| **Task Manager (Middle Load)** | **Processes** | |
| | Member of Active Processes | 81 |
| | **CPU Performance** | |
| | CPU Utilization (%) | 100% |
| | CPU Speed (GHz) | 2.70 GHz |
| | No. Threads | 2820 |
| | No. Handles | 122766 |
| | **Memory Performance** | |
| | Total Memory (GB) | 8.0 GB |
| | Used Memory (GB) | 7.8 GB |
| | Memory Usage (%) | 73% |
| | In Use (Compressed) (GB) | 5.7 GB (312 MB) |
| | Committed Memory (GB) | 9.4/12.0 GB |
| | Cache Memory (GB) | 2.0 GB |
| **Task Manager (High Load)** | **Processes** | |
| | Member of Active Processes | 80 |
| | **CPU Performance** | |
| | CPU Utilization (%) | 100% |
| | CPU Speed (GHz) | 2.70 GHz |
| | No. Threads | 3089 |
| | No. Handles | 127601 |
| | **Memory Performance** | |
| | Total Memory (GB) | 8.0 GB |
| | Used Memory (GB) | 7.8 GB |
| | Memory Usage (%) | 88% |
| | In Use (Compressed) (GB) | 6.8 GB (361 MB) |
| | Committed Memory (GB) | 10.7/13.1 GB |
| | Cache Memory (GB) | 1013 |
| **MiniTool Partition Wizard (32 KB)** | Physical Disk | Disk 1 – Micron_2450_MTFDKBA256TFK |
| | **Stress Test** | |
| | Sequential Reading (MB/s) | 1547.51 MB/s |
| | Sequential Writing (MB/s) | 1105.87 MB/s |
| | Random Reading (MB/s) | 1095.71 MB/s |
| | Random Writing (MB/s) | 1047.04 MB/s |

### 3.4 Dell Inspiron 16 5260 0D03J8

| Tool | Result Details | |
|---|---|---|
| Model | Dell Inspiron 16 5260 0D03J8<br> | |
| CPU-Z | Motherboard | 0D03J8 |
| | CPU | Intel Core i7 1260P |
| | Clock speed | 1795.61 MHz |
| | Cache | L1 Cache: 448KB<br>L2 Cache: 9MB<br>L3 Cache: 18MB |
| | Memory | 16 GBytes |
| Geekbench | **Benchmark Score** | |
| | Single-Core | 2297 |
| | Multi-Core | 7345 |
| | **Performance Result** | |
| | File compression: | |
| | Single-Core | 2007(288.2 MB/sec) |
| | Multi-Core | 2999(430.7 MB/sec) |
| | Navigation: | |
| | Single-Core | 2219 (13.4 routes/sec) |
| | Multi-Core | 9437 (56.9 routes/sec) |
| | HTML 5 Browser | |
| | Single-Core | 2211 (45.3 pages/sec) |
| | Multi-Core | 7176 (146.9 pages/sec) |
| Task manager (Low Load) | **Processes** | |
| | Number of Active Processes | 82 |
| | **CPU Performance** | |
| | CPU Utilization (%) | 3 |
| | CPU Speed (GHz) | 1.40 |
| | No. Threads | 3722 |
| | No. Handles | 130229 |
| | **Memory Performance** | |
| | Total Memory (GB) | 15.7 |

| | | |
|---|---|---|
| | Used Memory (GB) | 8.6 |
| | Memory Usage (%) | 54 |
| | In Use (Compressed) (GB) | 0.615 |
| | Committed Memory (GB) | 11.0/19.2 |
| | Cache Memory (GB) | 4.8 |
| Task manager (Middle Load) | **Processes** | |
| | Number of Active Processes | 83 |
| | **CPU Performance** | |
| | CPU Utilization (%) | 7 |
| | CPU Speed (GHz) | 1.67 |
| | No. Threads | 4289 |
| | No. Handles | 140369 |
| | **Memory Performance** | |
| | Total Memory (GB) | 16.0 |
| | Used Memory (GB) | 9.9 |
| | Memory Usage (%) | 65 |
| | In Use (Compressed) (GB) | 0.732 |
| | Committed Memory (GB) | 12.7/19.2 |
| | Cache Memory (GB) | 5.3 |
| Task manager (High Load) | **Processes** | |
| | Number of Active Processes | 99 |
| | **CPU Performance** | |
| | CPU Utilization (%) | 15 |
| | CPU Speed (GHz) | 1.09 |
| | No. Threads | 5450 |
| | No. Handles | 183985 |
| | **Memory Performance** | |
| | Total Memory (GB) | 16.0 |
| | Used Memory (GB) | 13.4 |
| | Memory Usage (%) | 87 |
| | In Use (Compressed) (GB) | 0.94 |
| | Committed Memory (GB) | 18.5/21.2 |
| | Cache Memory (GB) | 2.1 |
| MiniTool Partition Wizard (32KB) | Physical Disk | NVMe KBG50ZNS512G NVMe KIOXIA 512GB |
| | **Stress Test** | |
| | Sequential Reading (MB/s) | 1794.15 |
| | Sequential Writing (MB/s) | 1255.91 |
| | Random Reading (MB/s) | 1755.8 |
| | Random Writing (MB/s) | 1277.83 |

## Comparison on Benchmarking Result

| | | | | |
|---|---|---|---|---|
| Model | Acer Aspire M5-481 PTG | VivoBook ASUSLaptop X515JA A516JA | VivoBook ASUSLaptop X415MA A416MA | Dell Inspiron 16 5260 0D03J8 |
| **CPU-Z** | | | | |
| CPU processor | Intel Core i5 3337U | Intel Core i3 1005G1 | Intel Celeron N4020 CPU 1.10GHz | Intel Core i7 1260P |
| Clock speed | 798.1 MHz | 1995.6 MHz | 795.64 MHz | 1296.83 MHz |
| Cache | L1 Cache: 128KB<br>L2 Cache: 512KB<br>L3 Cache: 3MB | L1 Cache: 160KB<br>L2 Cache: 1024KB<br>L3 Cache: 4MB | L1 Cache: 48KB<br>L2 Cache: 62KB<br>L3 Cache: 4MB | L1 Cache: 448KB<br>L2 Cache: 9MB<br>L3 Cache: 18MB |
| Memory | 10 GBytes | 12 GBytes | 8 GBytes | 16 GBytes |
| Mother board | MA40_HX | X515JA | X415MA | A00 |
| **GEEKBENCH** | | | | |
| **Benchmark Score** | | | | |
| Single-Core | 356 | 1244 | 329 | 2297 |
| Multi-Core | 717 | 2319 | 548 | 7345 |
| **Performance Result** | | | | |
| File Compression (Single-Core) | 324(46.5 MB/sec) | 1053 (151.2 MB/s) | 458 (65.8 MB/s) | 2007 (288.2 MB/s) |
| File Compression (Multi-Core) | 773(111.0 MB/sec) | 1275 (183.1 MB/s) | 496 (71.2 MB/s) | 2999 (430.7 MB/s) |
| Navigation (Single-Core) | 819 (4.93 MB/s) | 1135 (6.84 routes/sec) | 616 (3.71 routes/s) | 2219 (13.4 routes/s) |
| Navigation (Multi-Core) | 1730 (10.4 MB/s) | 2610 (15.70 routes/sec) | 1094 (6.59 routes/s) | 9437 (56.9 routes/s) |
| HTML 5 Browser (Single-Core) | 680 (13.9 pages/s) | 1313 (26.9 pages/s) | 560 (11.5 pages/s) | 2211 (45.3 pages/s) |
| HTML 5 Browser (Multi-Core) | 1169 | 2620 (53.6 pages/s) | 964 (19.7 pages/s) | 7176 |

| | | | | | |
|---|---|---|---|---|---|
| | | (23.9 pages/s) | | | (146.9 pages/s) |
| **TASK MANAGER** | | | | | |
| Low Load | **Processes** | | | | |
| | Number of Active Processes | 78 | 87 | 83 | 82 |
| | **CPU Performance** | | | | |
| | CPU Utilization (%) | 30 | 49 | 38 | 3 |
| | CPU Speed (GHz) | 2.17 | 1.41 | 1.31 | 1.40 |
| | No. Threads | 2241 | 3115 | 2479 | 3722 |
| | No. Handles | 81144 | 114410 | 110868 | 130229 |
| | **Memory Performance** | | | | |
| | Total Memory (GB) | 9.8 | 11.7 | 7.8 | 15.7 |
| | Used Memory (GB) | 3.7 | 7.3 | 4.9 | 8.6 |
| | Memory Usage (%) | 39 | 60 | 63 | 54 |
| | In Use (Compressed) (GB) | 0.341 | 0 | 0.307 | 0.615 |
| | Committed Memory (GB) | 4.2/11.3 | 7.9/21.1 | 8.3/12.0 | 11.0/19.2 |
| | Cache Memory (GB) | 5.7 | 4.1 | 2.3 | 4.8 |
| | **Processes** | | | | |
| | Number of Active Processes | 126 | 84 | 85 | 83 |
| | **CPU Performance** | | | | |
| | CPU Utilization (%) | 54 | 51 | 100 | 7 |

| | | | | | |
|---|---|---|---|---|---|
| Middle Load | CPU Speed (GHz) | 2.48 | 1.38 | 2.70 | 1.67 |
| | No. Threads | 2659 | 2626 | 2820 | 4289 |
| | No. Handles | 102358 | 106695 | 122766 | 140369 |
| | **Memory Performance** | | | | |
| | Total Memory (GB) | 9.8 | 11.7 | 7.8 | 16.0 |
| | Used Memory (GB) | 5.9 | 6.8 | 5.7 | 9.9 |
| | Memory Usage (%) | 60 | 63 | 71 | 65 |
| | In Use (Compressed) (GB) | 0.56 | 0 | 0.312 | 0.732 |
| | Committed Memory (GB) | 7.9/11.3 | 7.1/21.1 | 9.4/12.0 | 12.7/19.2 |
| | Cache Memory (GB) | 3.5 | 4.8 | 2.0 | 5.3 |
| High Load | **Processes** | | | | |
| | Number of Active Processes | 138 | 87 | 85 | 99 |
| | **CPU Performance** | | | | |
| | CPU Utilization (%) | 100 | 100 | 100 | 15 |
| | CPU Speed (GHz) | 2.28 | 1.67 | 2.70 | 1.09 |
| | No. Threads | 3438 | 3023 | 3089 | 5450 |
| | No. Handles | 120358 | 105977 | 127601 | 183985 |
| | **Memory Performance** | | | | |
| | Total Memory (GB) | 9.8 | 11.7 | 7.8 | 16.0 |

| | | | | | |
|---|---|---|---|---|---|
| | Used Memory (GB) | 7.3 | 7.2 | 6.8 | 13.4 |
| | Memory Usage (%) | 76 | 60 | 87 | 87 |
| | In Use (Compressed) (GB) | 0.223 | 0 | .361 | 0.94 |
| | Committed Memory (GB) | 9.1/11.3 | 7.0/21.1 | 10.7/13.1 | 18.5/21.2 |
| | Cache Memory (GB) | 2.4 | 2.8 | 1.013 | 2.1 |
| **MiniTool Partition Wizard** | | | | | |
| Physical Disk | | KINGSTON SA400S37 480 ATA | SAMSUNG MZVLQ256HAJD-00000 | Micron_2450_MTFDKBA256TFK | NVMe KBG50ZNS512G NVMe KIOXIA 512 GB |
| Sequential reading (MB/s) | | 182.577 | 455.023 | 1547.51 | 1749.15 |
| Sequential Writing (MB/s) | | 451.567 | 100.017 | 1105.87 | 1255.91 |
| Random Reading (MB/s) | | 351.743 | 259.783 | 1095.71 | 1755.8 |
| Random writing (MB/s) | | 409.714 | 108.901 | 1047.04 | 1277.83 |

# CPU-Z

## Acer Aspire M5-481 PTG

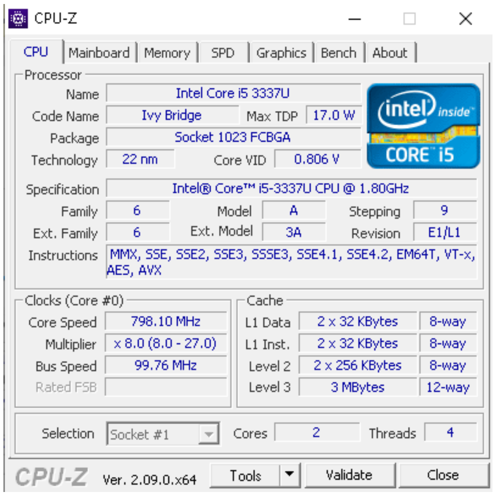



Figure A1: Example of CPU-Z output

## VivoBook ASUS Laptop X515JA A516

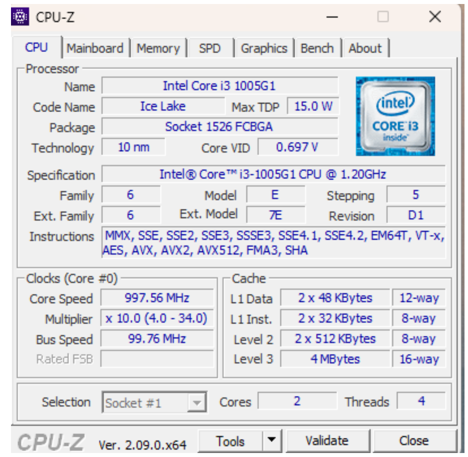



Figure A1: Example of CPU-Z outpu

## VivoBook ASUSLaptop X415MA A416MA

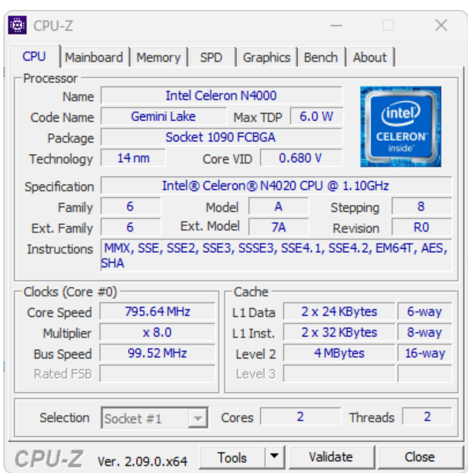



Figure A1: Example of CPU-Z output

## Dell Inspiron 16 52600D03J8

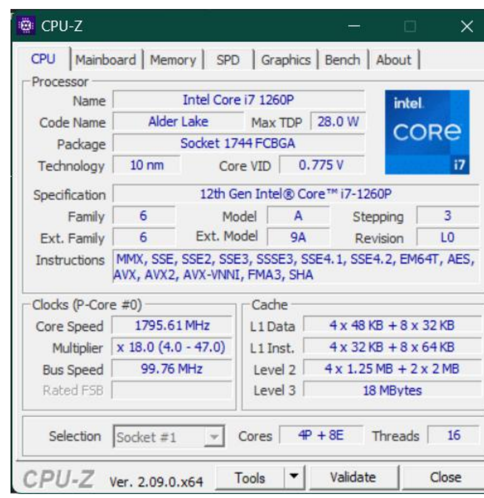
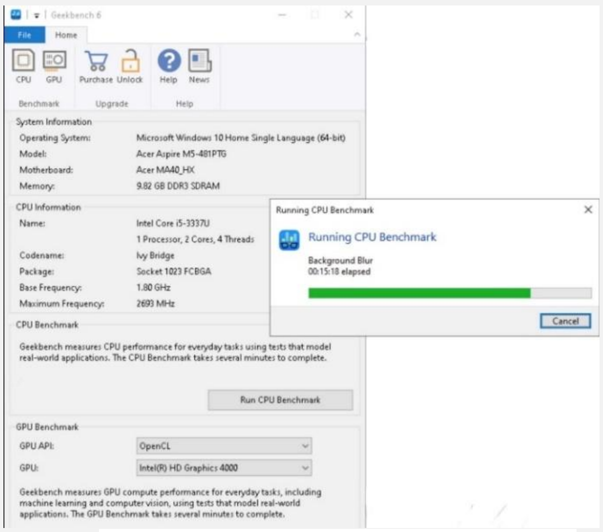



Figure A1: Example of CPU-Z outpu

# GEEKBENCH

## Acer Aspire M5-481 PTG

Figure A2: Example of Geekbench output




| Single-Core Performance | | Multi-Core Performance | |
|---|---|---|---|
| Single-Core Score | 356 | Single-Core Score | 717 |
| File Compression | 324<br>46.5 MB/sec | File Compression | 773<br>111.0 MB/sec |
| Navigation | 819<br>4.93 routes/sec | Navigation | 1730<br>10.4 routes/sec |
| HTML5 Browser | 680<br>13.9 pages/sec | HTML5 Browser | 1169<br>23.9 pages/sec |
| PDF Renderer | 610<br>14.1 Mpixels/sec | PDF Renderer | 1385<br>31.9 Mpixels/sec |
| Photo Library | 158<br>2.15 images/sec | Photo Library | 298<br>4.04 images/sec |
| Clang | 590<br>2.91 Klines/sec | Clang | 1143<br>5.63 Klines/sec |
| Text Processing | 569<br>45.5 pages/sec | Text Processing | 696<br>55.7 pages/sec |
| Asset Compression | 484<br>15.0 MB/sec | Asset Compression | 1325<br>41.1 MB/sec |
| Object Detection | 39<br>1.17 images/sec | Object Detection | 77<br>2.30 images/sec |
| Background Blur | 342<br>1.42 images/sec | Background Blur | 544<br>2.25 images/sec |
| Horizon Detection | 388<br>12.1 Mpixels/sec | Horizon Detection | 1096<br>34.1 Mpixels/sec |
| Object Remover | 372<br>28.6 Mpixels/sec | Object Remover | 698<br>53.7 Mpixels/sec |
| HDR | 395<br>11.6 Mpixels/sec | HDR | 877<br>25.7 Mpixels/sec |
| Photo Filter | 250<br>2.49 images/sec | Photo Filter | 465<br>4.61 images/sec |
| Ray Tracer | 563<br>544.4 Kpixels/sec | Ray Tracer | 1284<br>1.24 Mpixels/sec |
| Structure from Motion | 199<br>6.30 Kpixels/sec | Structure from Motion | 417<br>13.2 Kpixels/sec |

Figure A3: CPU Load Stress Test (CPU Test)

## VivoBook ASUS Laptop X515JA A516

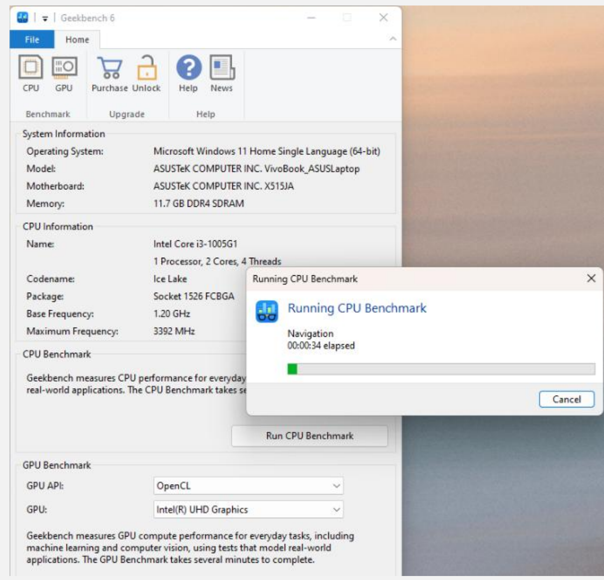



Figure B2: Example of Geekbench output

CPU Test

| Single-Core Performance | | Multi-Core Performance | |
|---|---|---|---|
| Single-Core Score | 1244 | Multi-Core Score | 2319 |
| File Compression | 1053<br>151.2MB/sec | File Compression | 1275<br>183.1 MB/sec |
| Navigation | 1135<br>6.84 routes/sec | Navigation | 2610<br>15.7 routes/sec |
| HTML5 Browser | 1313<br>26.9 pages/sec | HTML5 Browser | 2620<br>53.6 pages/sec |
| PDF Renderer | 1242<br>28.6 Mpixels/sec | PDF Renderer | 2908<br>67.1 Mpixels/sec |
| Photo Library | 1307<br>17.7 images/sec | Photo Library | 2620<br>35.6 images/sec |
| Clang | 1012<br>4.98 Klines/sec | Clang | 2411<br>11.9 Klines/sec |
| Text Processing | 1028<br>82.3 pages/sec | Text Processing | 1265<br>101.3 pages/sec |
| Asset Compression | 1286<br>39.8 MB/sec | Asset Compression | 2737<br>84.8 MB/sec |
| Object Detection | 1235<br>36.9 images/sec | Object Detection | 1790<br>53.6 images/sec |
| Background Blur | 1627<br>6.73 images/sec | Background Blur | 2648<br>11.0 images/sec |
| Horizon Detection | 1836<br>57.1 Mpixels/sec | Horizon Detection | 3664<br>114.0 Mpixels/sec |
| Object Remover | 867<br>66.7 Mpixels/sec | Object Remover | 1756<br>135.0 Mpixels/sec |
| HDR | 1327<br>38.9 Mpixels/sec | HDR | 2443<br>71.7 Mpixels/sec |
| Photo Filter | 1599<br>15.9 images/sec | Photo Filter | 2810<br>27.9 images/sec |
| Ray Tracer | 1295<br>1.25 Mpixels/sec | Ray Tracer | 2931<br>2.84 Mpixels/sec |
| Structure from Motion | 1401<br>44.4 Kpixels/sec | Structure from Motion | 2756<br>87.3 Kpixels/sec |

Figure B3: CPU Load Stress Test (CPU Test)

| VivoBook ASUSLaptop X415MA A416MA | Dell Inspiron 16 52600D03J8 |
|---|---|

## VivoBook ASUSLaptop X415MA A416MA

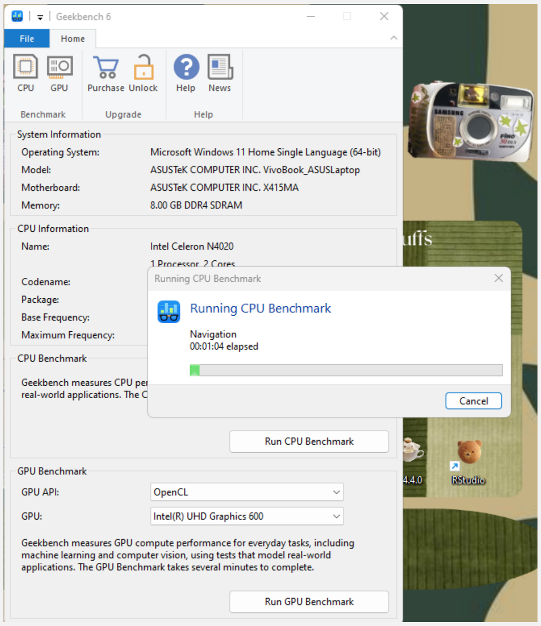



Figure C2: Example of Geekbench output

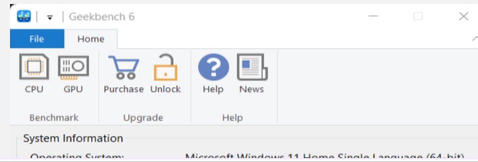
| Single-Core Performance | | Multi-Core Performance | |
|---|---|---|---|
| Single-Core Score | 329 | Multi-Core Score | 548 |
| File Compression | 458<br>65.8MB/sec | File Compression | 496<br>71.2 MB/sec |
| Navigation | 616<br>3.71 routes/sec | Navigation | 1094<br>6.59 routes/sec |
| HTML5 Browser | 560<br>11.5 pages/sec | HTML5 Browser | 964<br>19.7 pages/sec |
| PDF Renderer | 564<br>13.0 Mpixels/sec | PDF Renderer | 1090<br>25.1 Mpixels/sec |
| Photo Library | 151<br>2.05 images/sec | Photo Library | 260<br>3.53 images/sec |
| Clang | 555<br>2.73 Klines/sec | Clang | 941<br>4.63 Klines/sec |
| Text Processing | 536<br>42.9 pages/sec | Text Processing | 638<br>51.1 pages/sec |
| Asset Compression | 537<br>16.7 MB/sec | Asset Compression | 1009<br>31.3 MB/sec |
| Object Detection | 37<br>1.12 images/sec | Object Detection | 48<br>1.44 images/sec |
| Background Blur | 241<br>1.00 images/sec | Background Blur | 461<br>1.91 images/sec |
| Horizon Detection | 446<br>13.9 Mpixels/sec | Horizon Detection | 768<br>23.9 Mpixels/sec |
| Object Remover | 319<br>24.5 Mpixels/sec | Object Remover | 601<br>46.2 Mpixels/sec |
| HDR | 361<br>10.6 Mpixels/sec | HDR | 647<br>19.0 Mpixels/sec |
| Photo Filter | 195<br>1.93 images/sec | Photo Filter | 381<br>3.78 images/sec |
| Ray Tracer | 442<br>427.6 Kpixels/sec | Ray Tracer | 848<br>820.8 Kpixels/sec |
| Structure from Motion | 1401<br>5.63 Kpixels/sec | Structure from Motion | 351<br>11.1 Kpixels/sec |

Figure C3: CPU Load Stress Test (CPU Test)

Figure C3: CPU Load Stress Test (CPU Test)

## Dell Inspiron 16 52600D03J8

Figure

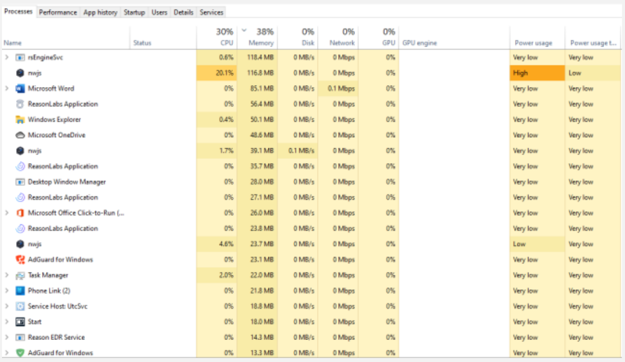
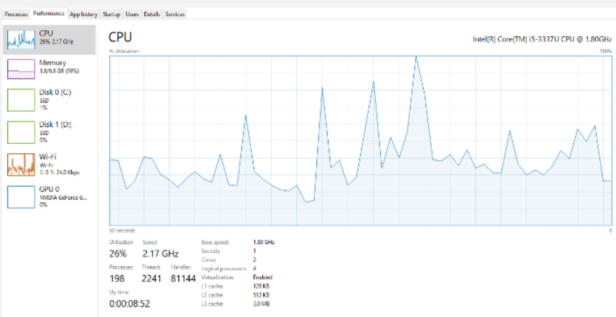
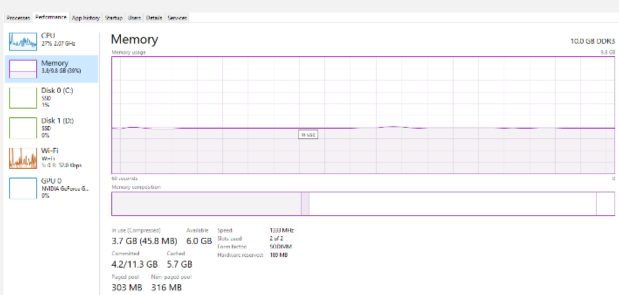
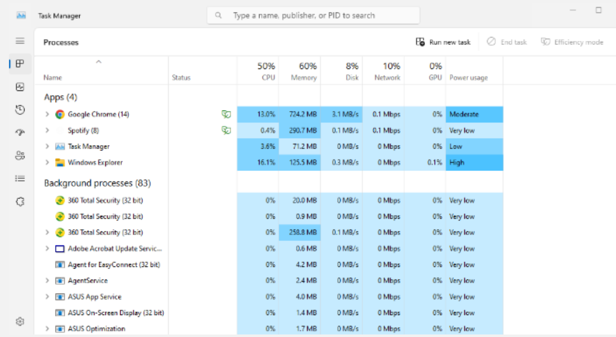
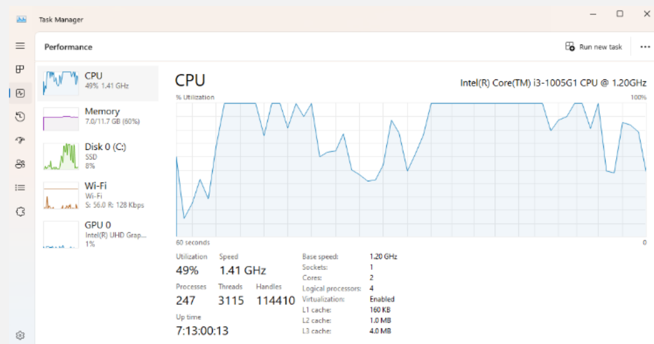
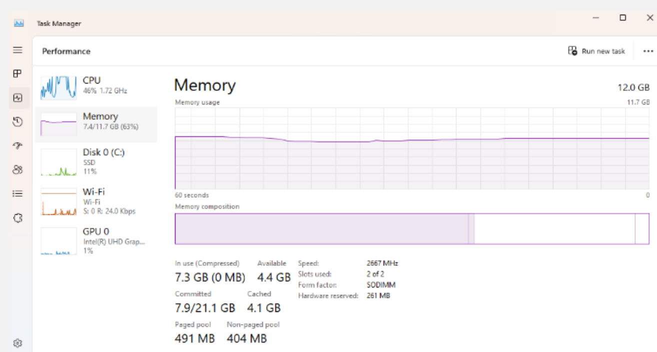



D2
Ge

| Single-Core Performance | | Multi-Core Performance | |
|---|---|---|---|
| Single-Core Score | 2297 | Multi-Core Score | 7345 |
| File Compression | 2007<br>288.2MB/sec | File Compression | 2999<br>430.7 MB/sec |
| Navigation | 2219<br>13.4 routes/sec | Navigation | 9437<br>56.9 routes/sec |
| HTML5 Browser | 2211<br>45.3 pages/sec | HTML5 Browser | 7176<br>146.9 pages/sec |
| PDF Renderer | 2288<br>52.8 Mpixels/sec | PDF Renderer | 9105<br>210.0 Mpixels/sec |
| Photo Library | 2305<br>31.3 images/sec | Photo Library | 8895<br>120.7 images/sec |
| Clang | 2224<br>11.0 Klines/sec | Clang | 12455<br>61.3 Klines/sec |
| Text Processing | 2189<br>175.3 pages/sec | Text Processing | 2472<br>198.0 pages/sec |
| Asset Compression | 2253<br>69.8 MB/sec | Asset Compression | 12993<br>402.6 MB/sec |
| Object Detection | 2189<br>65.5 images/sec | Object Detection | 4550<br>136.2 images/sec |
| Background Blur | 2903<br>12.0 images/sec | Background Blur | 8877<br>36.7 images/sec |
| Horizon Detection | 2703<br>84.1 Mpixels/sec | Horizon Detection | 9307<br>289.6 Mpixels/sec |
| Object Remover | 2122<br>163.2 Mpixels/sec | Object Remover | 9416<br>724.0 Mpixels/sec |
| HDR | 2261<br>66.3 Mpixels/sec | HDR | 7859<br>230.6 Mpixels/sec |
| Photo Filter | 2684<br>26.6 images/sec | Photo Filter | 6114<br>60.7 images/sec |
| Ray Tracer | 2246<br>2.17 Mpixels/sec | Ray Tracer | 9844<br>9.53 Mpixels/sec |
| Structure from Motion | 2479<br>78.5 Kpixels/sec | Structure from Motion | 9006<br>285.2 Kpixels/sec |

Figure D3: CPU Load Stress Test (CPU Test)

# Task Manager

| **Low Load** | |
|---|---|
| **Acer Aspire M5-481 PTG** | VivoBook ASUS Laptop X515JA A516 |
| <br>Figure A4.1: Process versus resources usage (CPU, Memory, Disk, Network, GPU)<br><br>Figure A4.2: Process versus resources usage (CPU)<br><br>Figure A4.3: Process versus resources usage (Memory) | <br>Figure B4.1: Process versus resources usage (CPU, Memory, Disk, Network, GPU<br><br>Figure B4.2: Process versus resources us (CPU)<br><br>Figure B4.3: Process versus resources us (Memory) |
| **VivoBook ASUSLaptop X415MA A416MA** | Dell Inspiron 16 52600D03J8 |

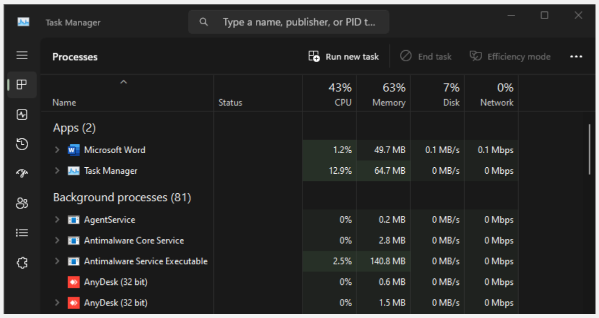


Figure C4.1: Process versus resources usage (CPU, Memory, Disk, Network, GPU)

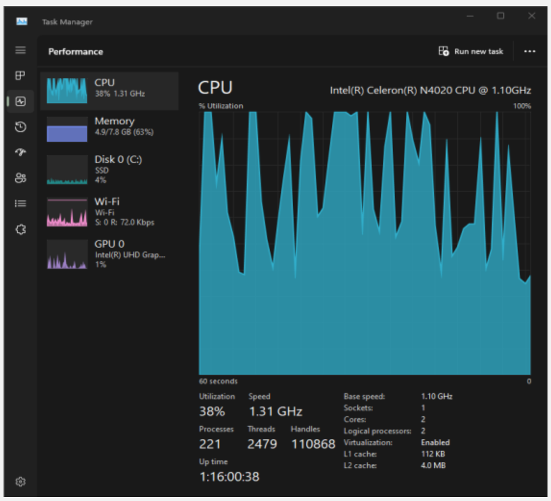


Figure C4.2: Process versus resources usage (CPU)

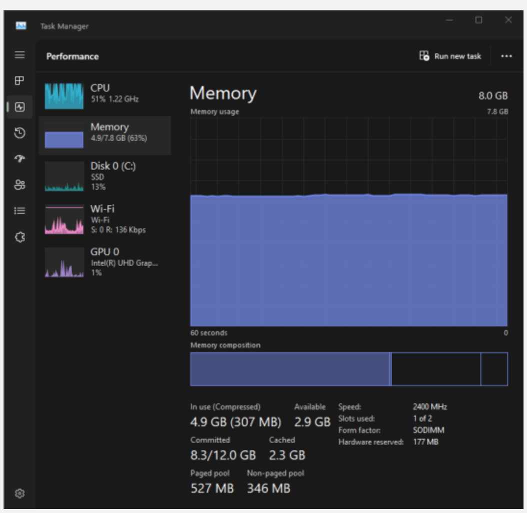


Figure C4.3: Process versus resources usage (Memory)

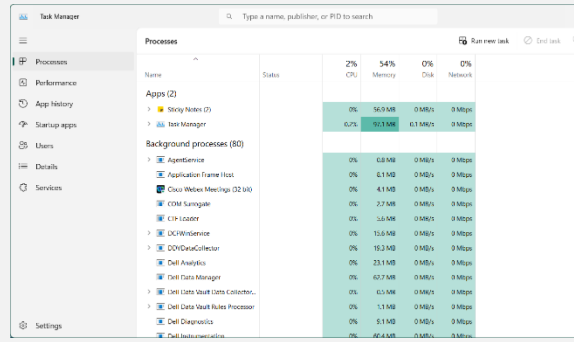


Figure D4.1: Process versus resources usag
(CPU, Memory, Disk, Network)

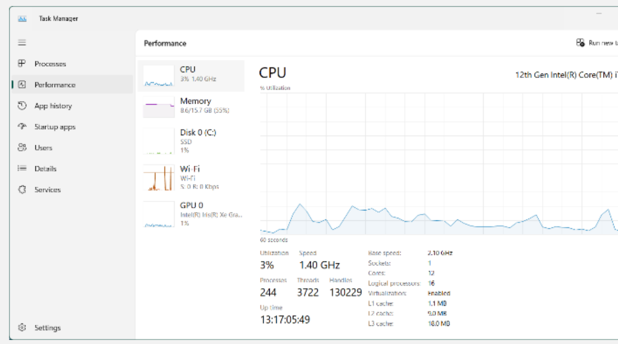


Figure D4.2: Process versus resources usage (C

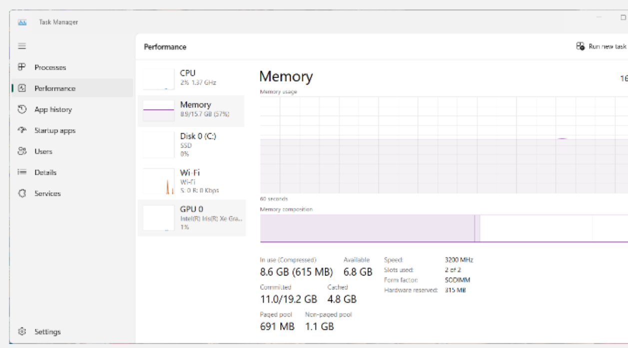


Figure D4.3:  Process versus resources usage (M

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<tr><th>Acer Aspire M5-481 PTG</th><th>VivoBook ASUS Laptop X515JA A516</th></tr>
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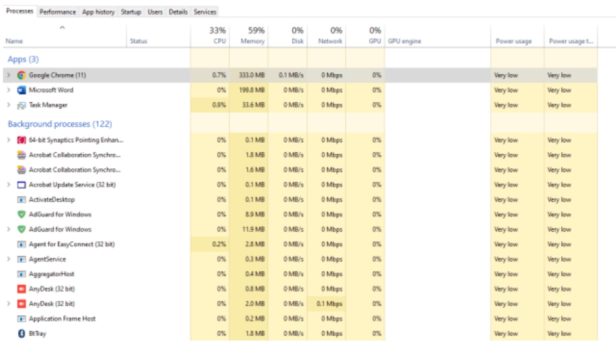

Figure A5.1: Process versus resources usage (CPU, Memory, Disk, Network, GPU)

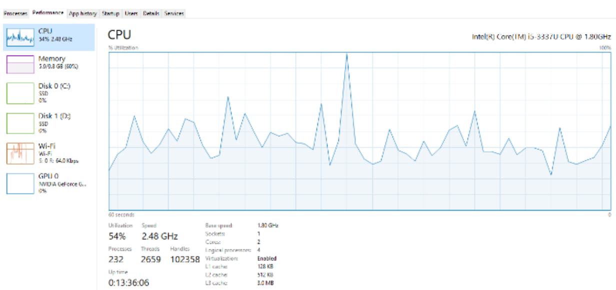

Figure A5.2: Process versus resources usage (CPU)

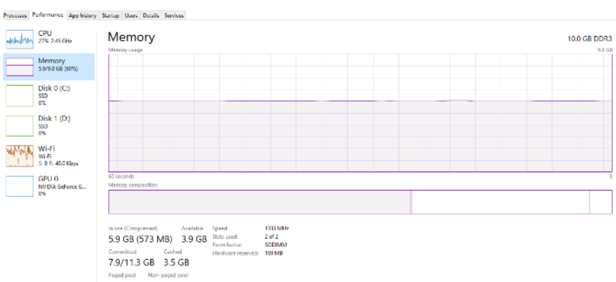

Figure A5.3: Process versus resources usage (Memory)
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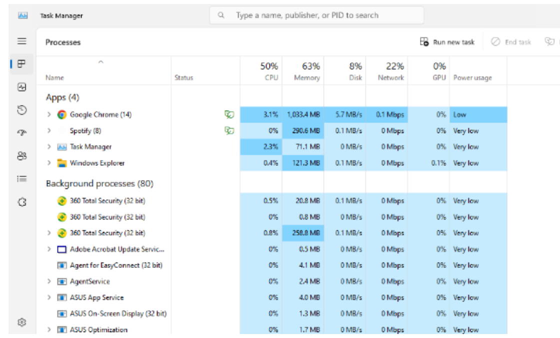

Figure B5.1: Process versus resources us
(CPU, Memory, Disk, Network, GPU

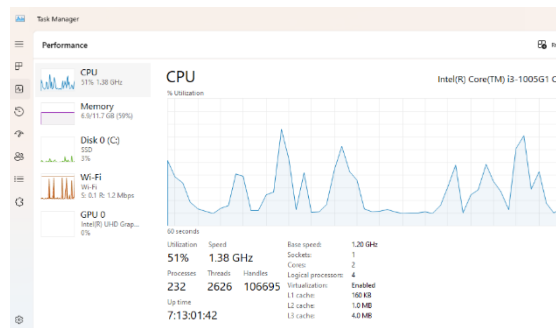

Figure B5.2: Process versus resources usage

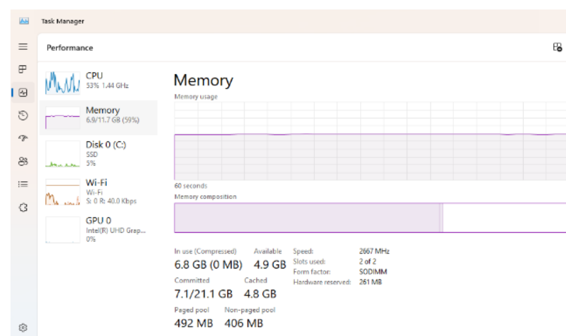

Figure B5.3: Process versus resources usage (
</td></tr>
</table>

| VivoBook ASUSLaptop X415MA A416MA | Dell Inspiron 16 52600D03J8 |
| --- | --- |

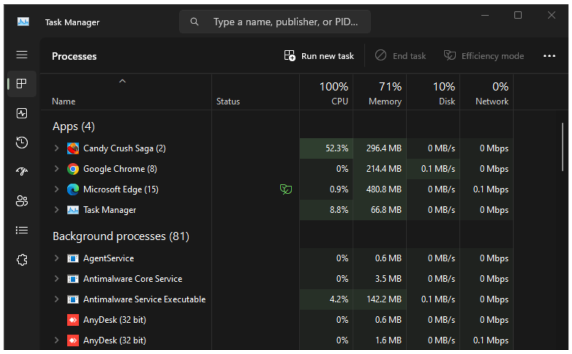


Figure C5.1: Process versus resources usage (CPU, Memory, Disk, Network, GPU)

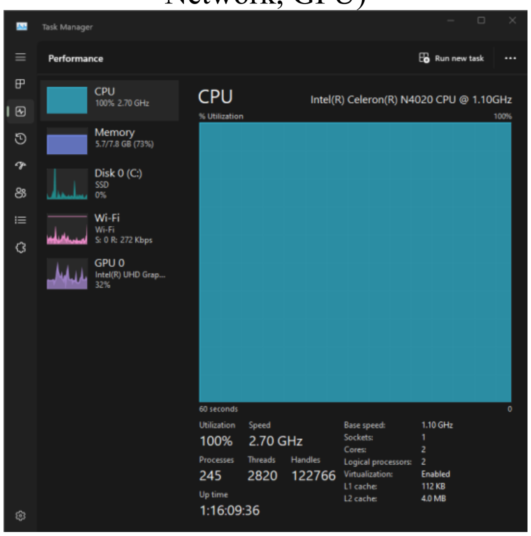


Figure C5.2: Process versus resources usage (CPU)

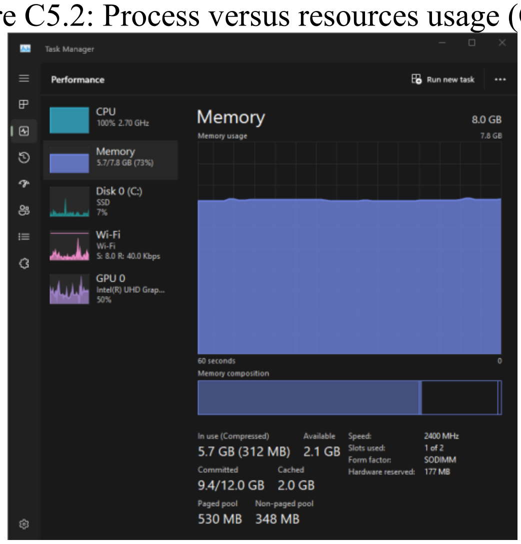


Figure C5.3: Process versus resources usage (Memory)

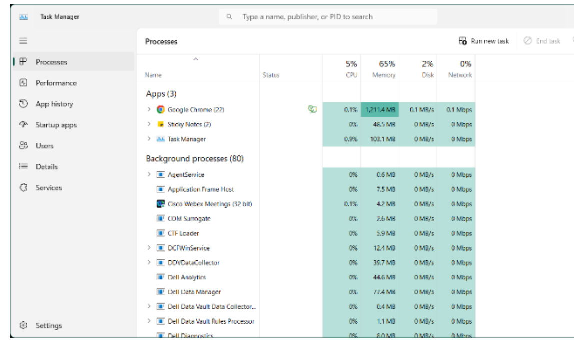


Figure D5.1: Process versus resources us (CPU, Memory, Disk, Network)

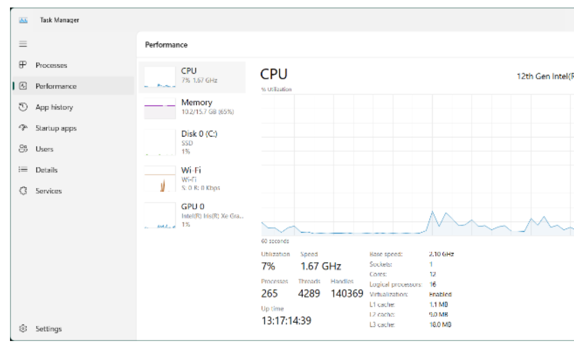


Figure D5.2: Process versus resources usage

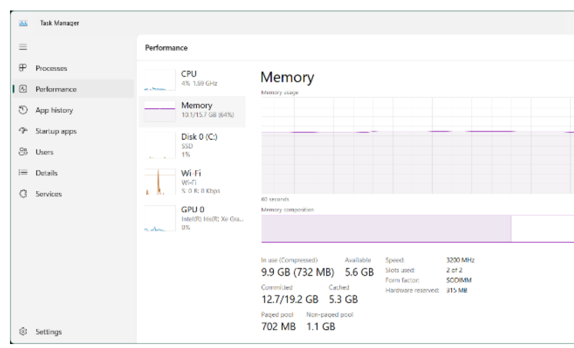


Figure D5.3: Process versus resources usage (

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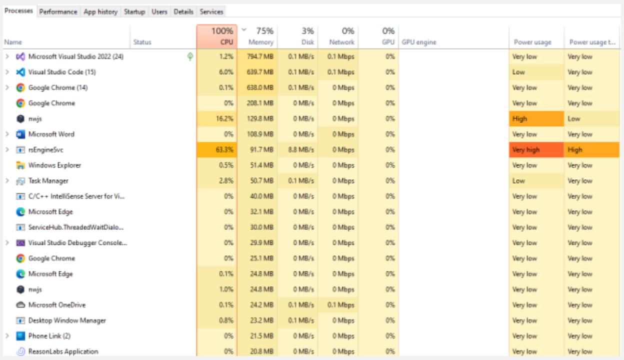
Figure A6.1: Process versus resources usage (CPU, Memory, Disk, Network, GPU)
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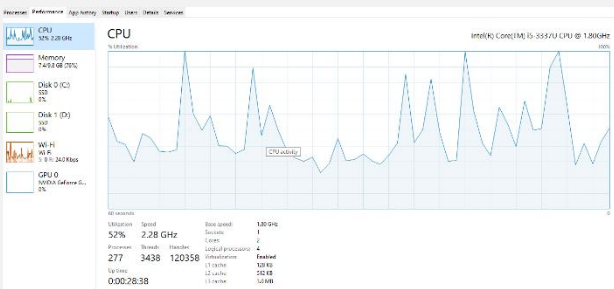
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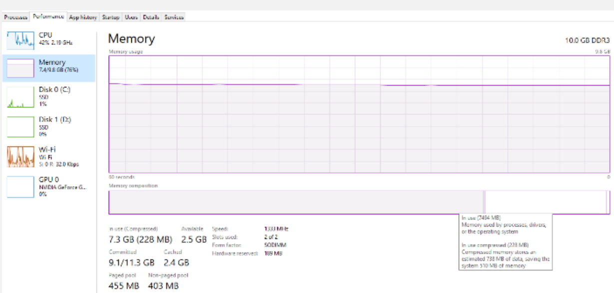
Figure A6.3: Process versus resources usage (Memory)
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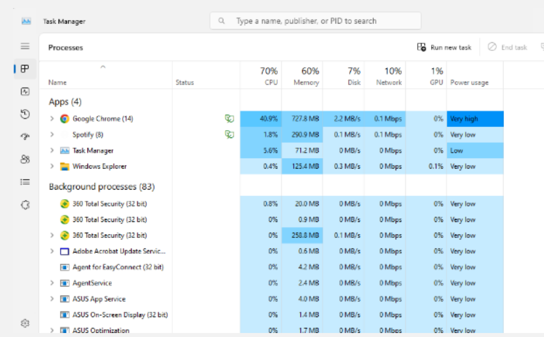
Figure B6.1: Process versus resources us (CPU, Memory, Disk, Network, GPU
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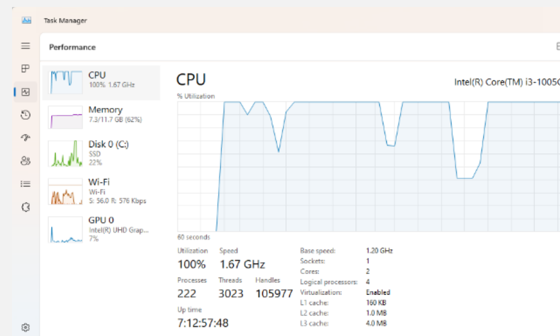
Figure B6.2: Process versus resources usage
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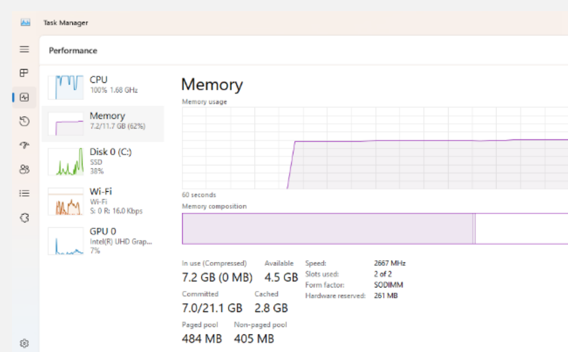


Figure B6.3: Process versus resources usage (
</td></tr>
</table>

| VivoBook ASUSLaptop X415MA A416MA | Dell Inspiron 16 52600D03J8 |
| --- | --- |

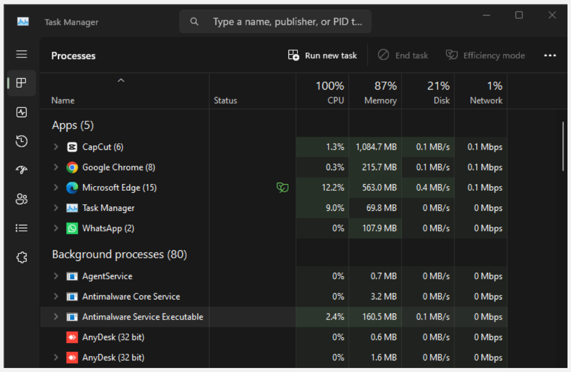



Figure C6.1: Process versus resources usage (CPU, Memory, Disk, Network, GPU)

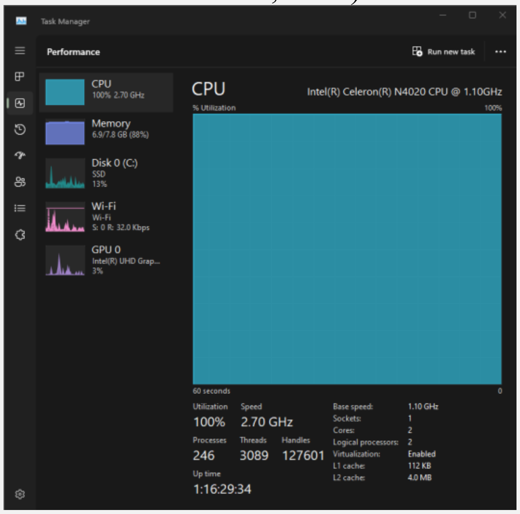



Figure C6.2: Process versus resources usage (CPU)

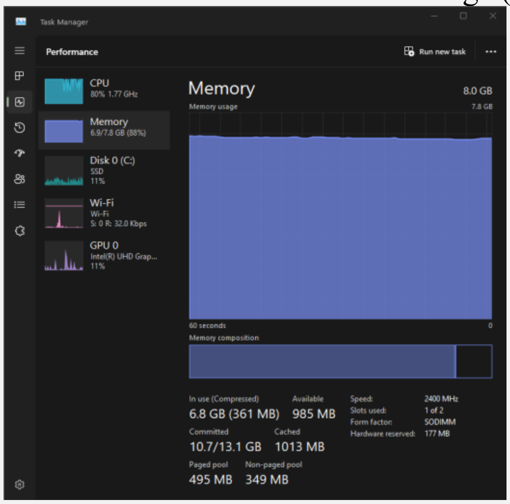



Figure C6.3: Process versus resources usage (Memory)

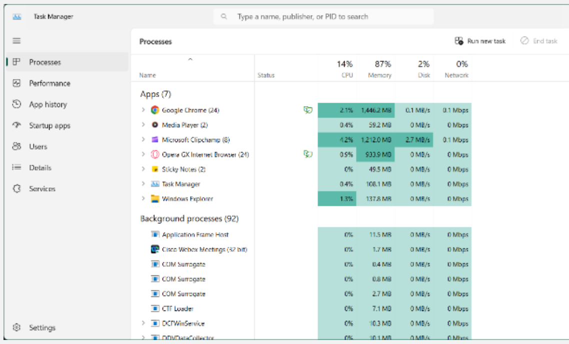



Figure D6.1: Process versus resources us
(CPU, Memory, Disk, Network, GPU

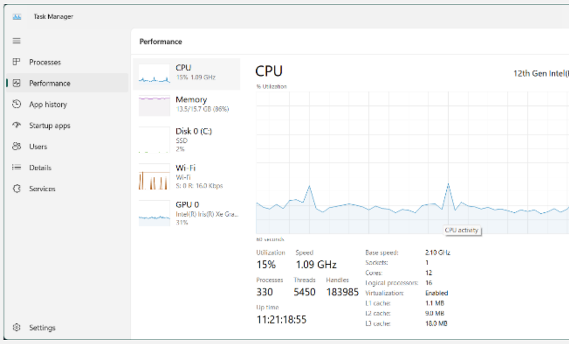



Figure D6.2: Process versus resources usage

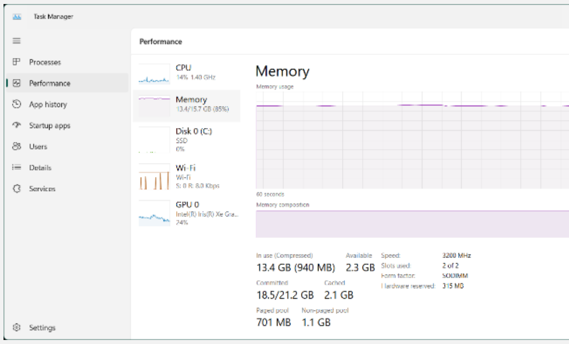
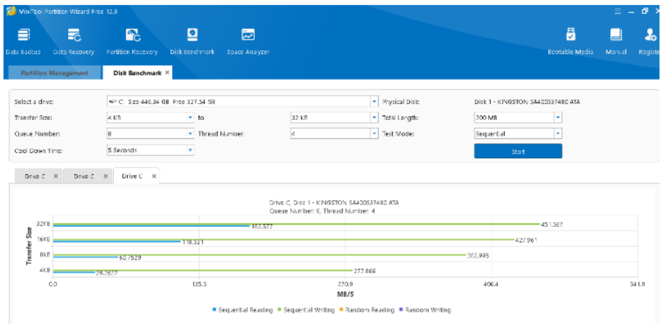
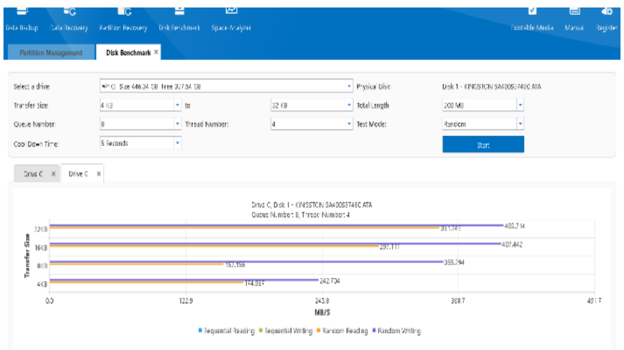
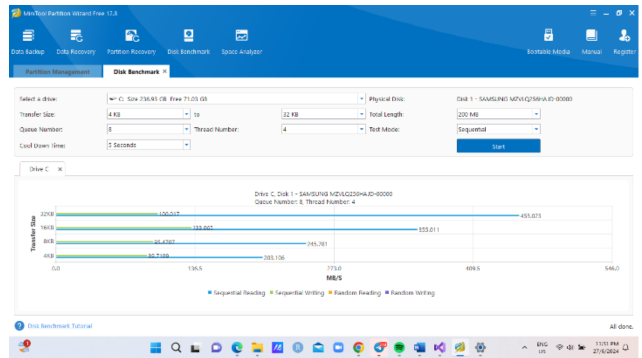
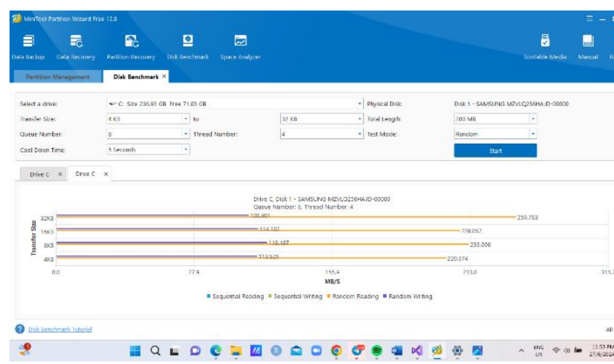



Figure D6.3: Process versus resources usage (

# MINITOOL PARTITION WIZARD

| Acer Aspire M5-481 PTG | VivoBook ASUS Laptop X515JA A516 |
| --- | --- |
|  Figure A7.1: Sequential mode  B7.2: mode Figure Random |  Figure B7.1: Sequential mode  |
| **VivoBook ASUSLaptop X415MA A416MA** | **Dell Inspiron 16 52600D03J8** |

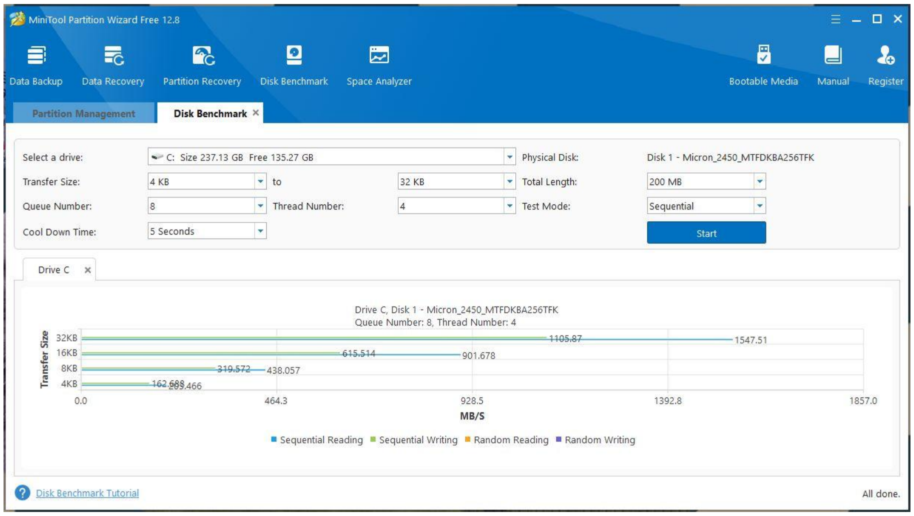



Figure C7.1: Sequential Mode

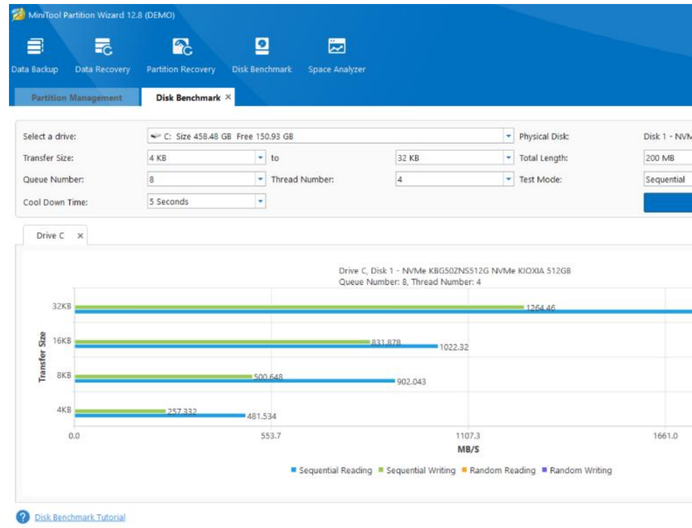



Figure D7.1: Sequential mode

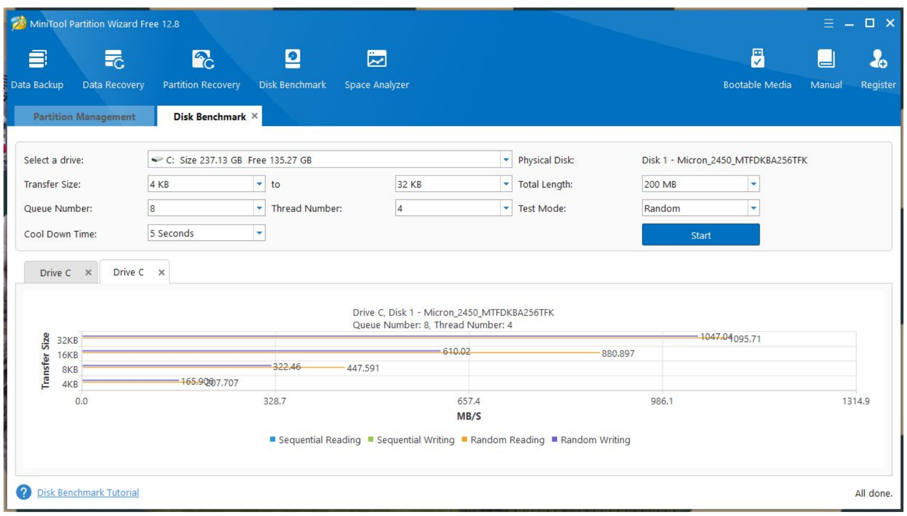



Figure C7.2: Random Mode

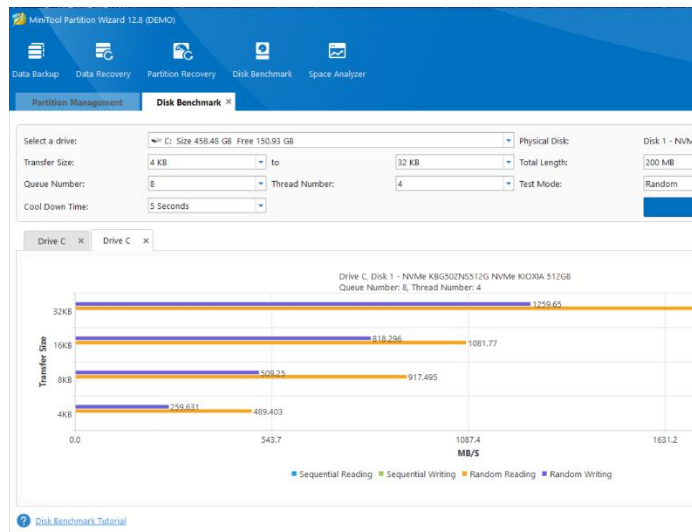



Figure D7.2: Random mode

# 4.0 Analysis, Comparison, Discussion

After running the benchmark test on four CPUs, we get the required performance data. Using the obtained data, we can obtain detailed information about CPU performance.

**CPU-Z**

| Model | Acer Aspire M5-481 PTG | VivoBook ASUS Laptop X515JA A516JA | VivoBook ASUS Laptop X415MA A416MA | Dell Inspiron 16 5260 0D03J8 |
|---|---|---|---|---|
| CPU processor | Intel Core i5 3337U | Intel Core i3 1005G1 | Intel Celeron N4020 CPU 1.10GHz | Intel Core i7 1260P |
| Clock speed | 798.1 MHz | 1995.6 MHz | 795.64 MHz | 1296.83 MHz |
| Cache | L1 Cache: 128KB<br>L2 Cache: 512KB<br>L3 Cache: 3MB | L1 Cache: 160KB<br>L2 Cache: 1024KB<br>L3 Cache: 4MB | L1 Cache: 48KB<br>L2 Cache: 62KB<br>L3 Cache: 4MB | L1 Cache: 448KB<br>L2 Cache: 9MB<br>L3 Cache: 18MB |
| Memory | 10 GBytes | 12 GBytes | 8 GBytes | 16 GBytes |
| Mother board | MA40_HX | X515JA | X415MA | A00 |

In this analysis, we compare the CPU performance of four different laptop models using CPU-Z benchmarking results. The laptops evaluated are the Acer Aspire M5-481PTG, VivoBook ASUS Laptop X515JA A516JA, VivoBook ASUS Laptop X415MA A416MA, and Dell Inspiron 16 5260 0D03J8. The analysis focuses on CPU processor (clock frequency), cache, memory, and motherboard.

In term of CPU processor, Acer Aspire M5-481PTG features an Intel Core i5 3337U processor running at a clock speed of 798.1 MHz. As an older generation CPU, its relatively low clock speed impacts its performance, especially when handling modern applications and multitasking. In contrast, the VivoBook ASUS Laptop X515JA A516JA, with its Intel Core i3 1005G1 processor operating at 1995.6 MHz, benefits from a higher clock speed, enabling it to process more instructions per second and perform better across various tasks. Similarly, the VivoBook ASUS Laptop X415MA A416MA, powered by an Intel Celeron N4020 CPU with a clock speed of 795.64 MHz, shares a similar clock speed with the Acer Aspire but is limited by its Celeron architecture, making it suitable only for basic computing tasks. On the other hand, the Dell Inspiron 16 5260 0D03J8, equipped with an Intel Core i7 1260P processor clocked at 1296.83 MHz, benefits from advanced architecture and additional cores, which significantly enhance its overall performance despite not having the highest clock speed in this comparison.

In term of cache, Acer Aspire M5-481PTG features a cache configuration of L1: 128KB, L2: 512KB, and L3: 3MB. These smaller cache sizes result in slower data retrieval and processing times compared to newer models. In contrast, the VivoBook ASUS Laptop X515JA A516JA offers larger cache sizes of L1: 160KB, L2: 1024KB, and L3: 4MB, enabling faster access to frequently used data and improved overall performance, particularly in multitasking scenarios. The VivoBook ASUS Laptop X415MA A416MA, with its minimal cache sizes of L1: 48KB, L2: 62KB, and L3: 4MB, is constrained in performance, especially in data-heavy applications. On the other hand, the Dell Inspiron 16 5260 0D03J8 stands out with the largest cache sizes of L1: 448KB, L2: 9MB, and L3: 18MB. These substantial cache sizes significantly enhance data processing speed and system responsiveness, making it ideal for high-performance tasks.

In term of memory, Acer Aspire M5-481PTG, equipped with 10 GB of RAM, provides a moderate capacity suitable for basic multitasking and everyday computing tasks, though it may struggle with more demanding applications. In contrast, the VivoBook ASUS Laptop X515JA A516JA, with 12 GB of RAM, offers better multitasking capability and smoother performance in memory-intensive applications compared to the Acer Aspire. The VivoBook ASUS Laptop X415MA A416MA, featuring 8 GB of RAM, has the lowest memory capacity among the models compared, which limits its ability to handle multiple applications simultaneously. On the other hand, the Dell Inspiron 16 5260 0D03J8 stands out with 16 GB of RAM, excelling in efficiently handling multiple applications and large files, thus offering the best performance for demanding tasks among the four models.

In term of motherboard, Acer Aspire M5-481PTG features the MA40_HX motherboard, which supports the older Intel Core i5 3337U processor. Its outdated design limits compatibility with newer hardware and technologies, reducing its potential for upgrades. In contrast, the VivoBook ASUS Laptop X515JA A516JA utilizes the X515JA motherboard, which supports the Intel Core i3 1005G1 processor. This modern design offers better compatibility with current hardware and features, making it more future-proof. The VivoBook ASUS Laptop X415MA A416MA comes with the X415MA motherboard, which supports the entry-level Intel Celeron N4020 CPU. This basic design is suitable for low-end computing tasks but offers limited performance and upgrade options. On the other hand, the Dell Inspiron 16 5260 0D03J8 is equipped with the A00 motherboard, which supports the high-performance Intel Core i7 1260P processor. This motherboard provides the most advanced features and compatibility, offering a robust platform for future upgrades and high-end computing tasks.

Based on their CPU features, these four laptops' comparison reveals significant performance variations. The greatest option for future-proofing and high-performance work is the Dell Inspiron 16 5260 0D03J8, which stands out for its superior processor, larger cache sizes, enough RAM, and modern motherboard. With a contemporary processor and respectable RAM, the VivoBook ASUS Laptop X515JA A516JA provides a balanced performance that is ideal for daily computing and light multitasking. With its mediocre features and outdated processor, the Acer Aspire M5-481PTG performs well for simple tasks but suffers with intensive apps. Finally, the VivoBook ASUS Laptop X415MA A416MA is best suited for basic computer tasks due to its entry-level processor and small RAM.

**Geekbench**

| Laptop Model | CPU | Clock Speed | Single-Core Score | Multi-Core Score |
|---|---|---|---|---|
| **Acer Aspire M5-481PTG** | Intel Core i5 3337U | 798.1 MHz | 356 | 717 |
| **VivoBook ASUSLaptop X515JA A516JA** | Intel Core i3 1005G1 | 1995.6 MHz | 1244 | 2319 |
| **VivoBook ASUSLaptop X415MA A416MA** | Intel Celeron N4020 CPU 1.10GHz | 795.64 MHz | 329 | 548 |
| **Dell Inspiron 16 5260 0D03J8** | Intel Core i7 1260P | 1296.83 MHz | 2297 | 7345 |

The Acer Aspire M5-481PTG is powered by an Intel Core i5 3337U processor, which is relatively older than the others in our analysis. The VivoBook ASUSLaptop X515JA A516JA has an Intel Core i3 1005G1, which is a more current and efficient processor. The VivoBook ASUSLaptop X415MA A416MA is powered by an Intel Celeron N4020 CPU, which is optimized for simple work and low power consumption. Finally, the Dell Inspiron 16 5260 0D03J8 has the latest and most powerful CPU in the group, the Intel Core i7 1260P, which is noted for its strong performance in both single and multi-threaded applications.

Clock speed is another significant factor in affecting CPU performance. The Acer Aspire M5-481PTG has a clock speed of 798.1 MHz, which is significantly lower than the others. The VivoBook ASUSLaptop X515JA A516JA operates at 1995.6 MHz, which is a significant boost in speed. The VivoBook ASUSLaptop X415MA A416MA has an Intel Celeron N4020 processor with a clock speed of 795.64 MHz, which is little lower than the Acer but still within a similar range. The Dell Inspiron 16 5260 0D03J8, powered by an Intel Core i7 1260P processor, operates at 1296.83 MHz, efficiently balancing performance and power efficiency.

The single-core score is an important measure of a processor's ability to handle tasks that require single-threaded performance. The Acer Aspire M5-481PTG has a single-core score of 302, indicating an outdated architecture and slower clock speed. The VivoBook ASUSLaptop X515JA A516JA performs far better, with a single-core score of 1244, due to its latest Intel Core i3 1005G1 processor. The VivoBook ASUSLaptop X415MA A416MA has a single-core score of 329, which is slightly higher than the Acer but still illustrates the limits of the Intel Celeron N4020 processor. The Dell Inspiron 16 5260 0D03J8 leads the category with a single-core score of 2297, demonstrating the better capability of the Intel Core i7 1260P processor.

Whenever it involves multi-core performance, which is critical for activities that might use several threads, the disparities are further noticeable. The Acer Aspire M5-481PTG gets a multi-core score of 557, which is relatively low given its ancient dual-core processor. The VivoBook ASUSLaptop X515JA A516JA performs far better, with a multi-core score of 2319, demonstrating the advantages of its modern architecture and hyper-threading capabilities. The VivoBook ASUSLaptop X415MA A416MA gets a multi-core score of 548, which is lower than the VivoBook X515JA but equivalent to the Acer Aspire due to the Intel Celeron N4020's constraints. The Dell Inspiron 16 5260 0D03J8 leads the pack with an amazing multi-core score of 7345, highlighting the benefits of its sophisticated Intel Core i7 1260P processor, which has several cores and threads.

**Monitor Task Manager (Windows OS tools)**

| Laptop Model | Low Load CPU Utilization (%) | Low Load Memory Usage (GB) | Middle Load CPU Utilization (%) | Middle Load Memory Usage (GB) | High Load CPU Utilization (%) | High Load Memory Usage (GB) |
|---|---|---|---|---|---|---|
| Acer Aspire M5-481PTG | 30 | 39 | 54 | 60 | 100 | 76 |
| VivoBook ASUSLapto X515JA A516JA | 49 | 60.83 | 51 | 56.67 | 100 | 60 |
| VivoBook ASUSLapto X415MA A416MA | 38 | 97.5 | 100 | 97.5 | 100 | 97.5 |
| Dell Inspiron 16 5260 0D03J8 | 3 | 54 | 7 | 65 | 15 | 87 |

Based on the observation throughout the four types of computers that has undergone task manager with three different loads, we can observe that VivoBook ASUSLaptop X415MA A416MA has the overall highest CPU utilization and memory usage whereas Dell Inspiron 16 5260 0D03J8 has the overall lowest CPU utilization and memory usage.

For low load, the expected CPU utilization is between 30% to 50%. The active processes that are running for Acer Aspire M5-481PTG, VivoBook ASUSLaptop X515JA A516JA, VivoBook ASUSLaptop X415MA A416MA and Dell Inspiron 16 5260 0D03J8 are 78, 83, 81 and 82 respectively. Acer Aspire M5-481PTG, VivoBook ASUSLaptop X515JA A516JA and VivoBook ASUSLaptop X415MA A416MA have similar CPU utilization with slight difference that still fits into the expected CPU utilization. However, Dell Inspiron 16 5260 0D03J8 has a drastically different CPU utilization which is only 3% although it has similar number of active processes running. This shows that the Dell Inspiron 16 5260 0D03J8's CPU has high capacity. For memory usage, Acer Aspire M5-481PTG has the least memory used whereas VivoBook ASUSLaptop X415MA A416MA has the highest memory used which is almost 100%. VivoBook ASUSLaptop X515JA A516JA and Dell Inspiron 16 5260 0D03J8 have moderate memory usage percentage.

For middle load, the expected CPU utilization is between 50% to 60%. The active processes that are running for Acer Aspire M5-481PTG, VivoBook ASUSLaptop X515JA A516JA, VivoBook ASUSLaptop X415MA A416MA and Dell Inspiron 16 5260 0D03J8 are 126, 80, 81 and 83 respectively. Acer Aspire M5-481PTG and VivoBook ASUSLaptop X515JA A516JA have similar CPU utilization with slight difference that still fits into the expected CPU utilization. VivoBook ASUSLaptop X415MA A416MA and Dell Inspiron 16 5260 0D03J8 have drastic difference in value from the expected CPU utilization where VivoBook ASUSLaptop X415MA A416MA has 100% CPU utilization and Dell Inspiron 16 5260 0D03J8 has only 7% CPU utilization. Even though the number of active processes running are almost similar for VivoBook ASUSLaptop X515JA A516JA, VivoBook ASUSLaptop X415MA A416MA and Dell Inspiron 16 5260 0D03J8, all three of the laptops have completely different CPU utilization percentage. Acer Aspire M5-481PTG has comparatively higher number of active processes running but has the ideal CPU utilization percentage. For memory usage, VivoBook ASUSLaptop X515JA A516JA has the least memory usage whereas VivoBook ASUSLaptop X415MA A416MA has the highest memory used which is almost 100%. Acer Aspire M5-481PTG, VivoBook ASUSLaptop X515JA A516JA and Dell Inspiron 16 5260 0D03J8 have similar memory usage percentage.

For high load, the expected CPU utilization is above 70%. The active processes that are running for Acer Aspire M5-481PTG, VivoBook ASUSLaptop X515JA A516JA, VivoBook ASUSLaptop X415MA A416MA and Dell Inspiron 16 5260 0D03J8 are 138, 83, 80 and 99 respectively. Acer Aspire M5-481PTG, VivoBook ASUSLaptop X515JA A516JA and VivoBook ASUSLaptop X415MA A416MA have a 100% CPU utilization whereas Dell Inspiron 16 5260 0D03J8 only has 15% CPU utilization despite having higher number of active processes running than  VivoBook ASUSLaptop X515JA A516JA and VivoBook ASUSLaptop X415MA A416MA. For memory usage, VivoBook ASUSLaptop X515JA A516JA has the least memory usage whereas VivoBook ASUSLaptop X415MA A416MA has the highest memory used which is almost 100%. Acer Aspire M5-481PTG and Dell Inspiron 16 5260 0D03J8 have moderate memory usage percentage.

**MiniTool Partition Wizard**

| Benchmark | Computer Type | | | |
|---|---|---|---|---|
| | Acer Aspire M5-481 PTG | VivoBook ASUSLaptop X515JA A516JA | VivoBook ASUSLaptop X415MA A416MA | Dell Inspiration 16 5260 0D03J8 |
| **Physical Disk** | KINGSTON SA400S374 80 ATA | SAMSUNG MZVLQ256HAJD -00000 | Micron_2450_ MTFDKBA25 6TFK | NVMe KBG50ZNS51 2G NVMe KIOXIA 512 GB |
| **Stress Test** | | | | |
| **Sequential Reading (MB/s)** | 182.577 | 455.023 | 1547.51 | 1749.15 |
| **Sequential Writing (MB/s)** | 451.567 | 100.017 | 1105.87 | 1255.91 |
| **Random Reading (MB/s)** | 351.743 | 259.783 | 1095.71 | 1755.8 |
| **Random Writing (MB/s)** | 409.714 | 108.901 | 1047.04 | 1277.83 |

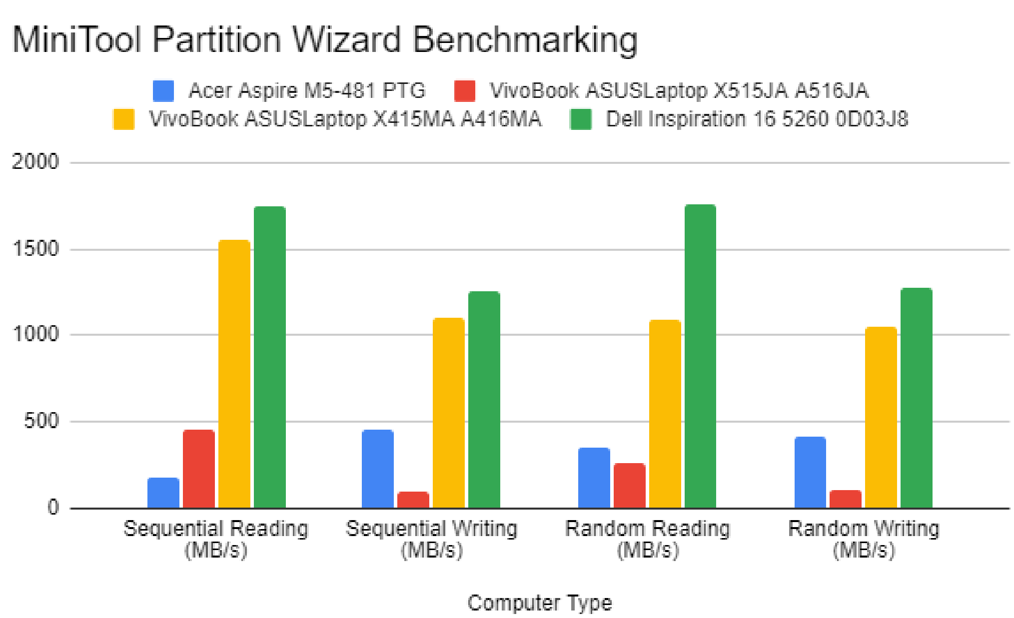



*Chart 4 MiniTool Partition Wizard Benchmarking*

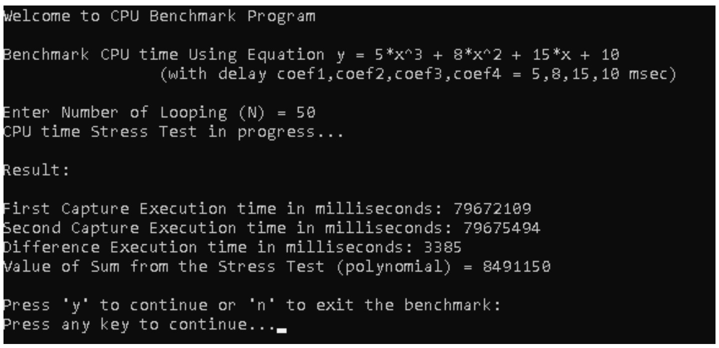
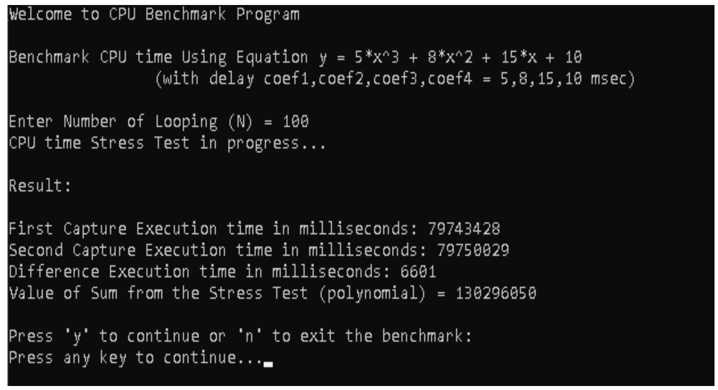
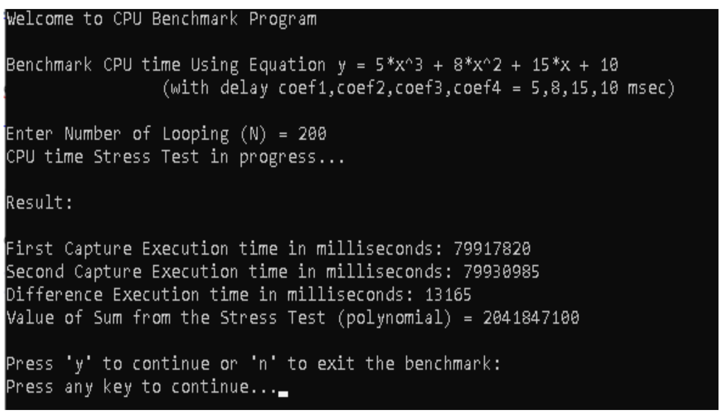
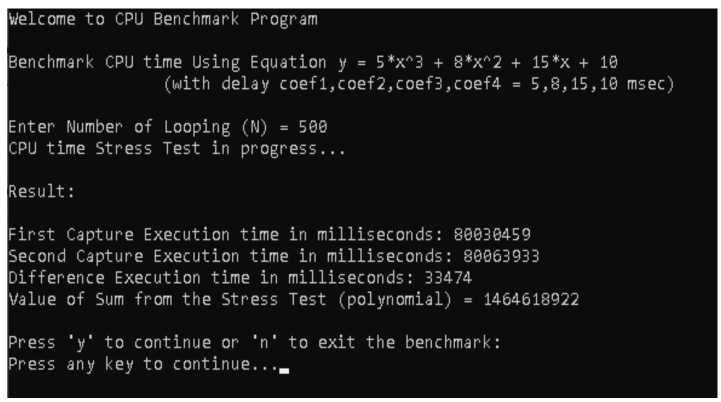
Based on the observation throughout the four types of computers performed the disk benchmark, we get to observed that for the sequential mode, Dell Inspiration 16 5260 0D03J8 has highest speed for sequential reading for 32KB which is 1749.15 MB/s compared to others three different types of computers. The second highest for the sequential mode test is VivoBook ASUSLaptop X415MA A416MA which indicates the sequential writing for 32KB with 1547.51 MB/s. Whereas VivoBook ASUSLaptop X515JA A516JA is the third position which got the result of 455.023 MB/s and the fourth is Acer Aspire M5-481 PTG with the result of 182.577 MB/s for the sequential reading in 32KB size. Having a faster sequential reading speed is critical for improving media streaming and file transfers on each laptop, since we all need high-quality video streaming and fast copying and transferring of large files. Thus, Dell Inspiration 16 5260 0D03J8 is the better option for the high quality of performance.

Next, for the sequential writing speed in the sequential test mode, we obtain that Dell Inspiration 16 5260 0D03J8 have the highest speed which is 1255.91 MB/s for 32KB transfer size among the others three laptops. VivoBook ASUSLaptop X415MA A416MA place second with the sequential writing of 1105.87 MB/s for 32KB transfer size. Acer Aspire M5-481 PTG has the slightly higher sequential writing speed compared to VivoBook ASUSLaptop X515JA A516JA for 32KB transfer size which each of them is 451.567 MB/s and 100.017 MB/s. High sequential writing speed is crucial for efficient backups operation, particularly when managing with the huge amounts of data. As a result, we can see that the Dell Inspiration 16 5260 0D03J8 is once again the best laptop option out of the three.

In addition, for the random test mode, Dell Inspiration 16 5260 0D03J8 achieve the highest random reading speed which is 1755.8 MB/s for 32KB transfer size. Meanwhile, the second higher for random reading disk benchmark is VivoBook ASUSLaptop X415MA A416MA with the speed of 1095.71 MB/s, and Acer Aspire M5-481 PTG obtain the third place with 351.743 MB/s for 32KB transfer size. VivoBook ASUSLaptop X515JA A516JA place last with the random reading speed value of 259.783 MB/s for 32KB transfer size. Excellent random reading rates maximize the performance of downloading applications that require accessing several small files dispersed within the disk. Dell Inspiration 16 5260 0D03J8 is the best laptop to have these opportunities compared to VivoBook ASUSLaptop X515JA A516JA, Acer Aspire M5-481 PTG, and VivoBook ASUSLaptop X415MA A416MA.

Lastly, for the random writing speed in the random test mode, Dell Inspiration 16 5260 0D03J8 have the highest speed which is 1277.83 MB/s for 32KB transferring size among the other three laptops. VivoBook ASUSLaptop X415MA A416MA rank second with the random writing speed of 1047.04 MB/s in 32KB transferring size. While the other two laptops, which are Acer Aspire M5-481 PTG and VivoBook ASUSLaptop X515JA A516JA obtain the random writing speed of 409.714 MB/s, and 108.901 MB/s in 32KB transfer size. Efficient random writing is critical for applications utilizing databases which frequently update relatively little amount of data. Dell Inspiration 16 5260 0D03J8 is the best choice to have a quick update on data compared to VivoBook ASUSLaptop X515JA A516JA, Acer Aspire M5-481 PTG, and VivoBook ASUSLaptop X415MA A416MA.

Thus, for the final System Benchmarks it is crucial to have higher chances at speed rates of both the sequential and random reading and writing tests because the performance of the laptop is largely based on the transfer rate speed. Consequently, the Dell Inspiration 16 5260 0D03J8 is an improved variant for users who seek attain higher speeds in sequential as well as in the random test modes.

| 50 Loop | 100 Loop |
| --- | --- |
| <br>Welcome to CPU Benchmark Program<br><br>Benchmark CPU time Using Equation y = 5*x^3 + 8*x^2 + 15*x + 10<br>(with delay coef1,coef2,coef3,coef4 = 5,8,15,10 msec)<br><br>Enter Number of Looping (N) = 50<br>CPU time Stress Test in progress...<br><br>Result:<br><br>First Capture Execution time in milliseconds: 79672109<br>Second Capture Execution time in milliseconds: 79675494<br>Difference Execution time in milliseconds: 3385<br>Value of Sum from the Stress Test (polynomial) = 8491150<br><br>Press 'y' to continue or 'n' to exit the benchmark:<br>Press any key to continue... | <br>Welcome to CPU Benchmark Program<br><br>Benchmark CPU time Using Equation y = 5*x^3 + 8*x^2 + 15*x + 10<br>(with delay coef1,coef2,coef3,coef4 = 5,8,15,10 msec)<br><br>Enter Number of Looping (N) = 100<br>CPU time Stress Test in progress...<br><br>Result:<br><br>First Capture Execution time in milliseconds: 79743428<br>Second Capture Execution time in milliseconds: 79750029<br>Difference Execution time in milliseconds: 6601<br>Value of Sum from the Stress Test (polynomial) = 130296050<br><br>Press 'y' to continue or 'n' to exit the benchmark:<br>Press any key to continue... |
| 200 Loop | 500 Loop |
| <br>Welcome to CPU Benchmark Program<br><br>Benchmark CPU time Using Equation y = 5*x^3 + 8*x^2 + 15*x + 10<br>(with delay coef1,coef2,coef3,coef4 = 5,8,15,10 msec)<br><br>Enter Number of Looping (N) = 200<br>CPU time Stress Test in progress...<br><br>Result:<br><br>First Capture Execution time in milliseconds: 79917820<br>Second Capture Execution time in milliseconds: 79930985<br>Difference Execution time in milliseconds: 13165<br>Value of Sum from the Stress Test (polynomial) = 2041847100<br><br>Press 'y' to continue or 'n' to exit the benchmark:<br>Press any key to continue... | <br>Welcome to CPU Benchmark Program<br><br>Benchmark CPU time Using Equation y = 5*x^3 + 8*x^2 + 15*x + 10<br>(with delay coef1,coef2,coef3,coef4 = 5,8,15,10 msec)<br><br>Enter Number of Looping (N) = 500<br>CPU time Stress Test in progress...<br><br>Result:<br><br>First Capture Execution time in milliseconds: 80030459<br>Second Capture Execution time in milliseconds: 80063933<br>Difference Execution time in milliseconds: 33474<br>Value of Sum from the Stress Test (polynomial) = 1464618922<br><br>Press 'y' to continue or 'n' to exit the benchmark:<br>Press any key to continue... |
| 700 Loop | 1000 Loop |

```
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation y = 5*x^3 + 8*x^2 + 15*x + 10
               (with delay coef1,coef2,coef3,coef4 = 5,8,15,10 msec)

Enter Number of Looping (N) = 700
CPU time Stress Test in progress...

Result:

First Capture Execution time in milliseconds: 80148514
Second Capture Execution time in milliseconds: 80194331
Difference Execution time in milliseconds: 45817
Value of Sum from the Stress Test (polynomial) = 1255716630

Press 'y' to continue or 'n' to exit the benchmark:
Press any key to continue...
```

```
Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation y = 5*x^3 + 8*x^2 + 15*x + 10
               (with delay coef1,coef2,coef3,coef4 = 5,8,15,10 msec)

Enter Number of Looping (N) = 1000
CPU time Stress Test in progress...

Result:

First Capture Execution time in milliseconds: 80279441
Second Capture Execution time in milliseconds: 80345180
Difference Execution time in milliseconds: 65739
Value of Sum from the Stress Test (polynomial) = 1048985068

Press 'y' to continue or 'n' to exit the benchmark:
Press any key to continue...
```

| No of Loop (N) | Capture_msec (before) | Capture_msec (after) | CPU_msec (different) |
|---|---|---|---|
| 50 | 79672109 | 79675494 | 3385 |
| 100 | 79743428 | 79750029 | 6601 |
| 200 | 79917820 | 79930985 | 13165 |
| 500 | 80030459 | 80063933 | 33474 |
| 700 | 80148514 | 80194331 | 45817 |
| 1000 | 80279441 | 80345180 | 65739 |

## 5.0 Conclusion and Reflection

### Conclusion

This research aimed to assess the CPU performance of several laptops using a variety of testing tools and techniques. The major goal was to evaluate and compare CPU efficiency across various systems using programmes like Geekbench, CPU-Z, Windows Task Manager, and MiniTool Partition Wizard. Additionally, the project included the construction of assembly code to assess execution times for various loop iterations, which improved understanding of CPU capabilities.

The benchmarking tools gave complete information on system performance. Geekbench provided a comprehensive collection of benchmarks for evaluating overall system performance, revealing strengths and weaknesses across various computers. CPU-Z provides thorough information on CPU specifications and performance indicators, enabling accurate comparisons of different CPUs. Windows Task Manager tracked real-time CPU utilization, providing insight into how different tasks and loads effect CPU performance. MiniTool Partition Wizard evaluated storage performance, which indirectly influenced overall system performance, notably data retrieval and application launching times.

Performance comparisons indicated considerable disparities amongst the tested computers. The Dell Inspiron 16 5260 0D03J8 consistently beat rival laptops in both sequential and random read/write tests, making it the ideal choice for tasks that need fast data transfers. Assembly code tests revealed that the Dell Inspiron 16 had the shortest execution speeds for loop iterations, indicating greater processing performance. These findings highlight the significance of sequential and random read/write speeds in overall system performance, as they affect application loading, data retrieval, and system responsiveness. The Dell Inspiron 16 demonstrates efficient CPU performance, which leads to speedier task processing and better handling of complicated programmes.

In conclusion, the study achieved its goals by providing a complete comparative examination of CPU performance across different laptops using multiple benchmarking methods. The hands-on experience with benchmarking tools and the creation of assembly code for performance testing greatly enhanced my understanding of CPU activities and their impact on overall system efficiency. Through coordinated efforts, the team effectively communicated its findings, demonstrating the practical use of CPU benchmarks in evaluating and enhancing computer performance.

## Reflection

**CHOH JING YI -** In summary, I have learned a lot from this project, which has improved my comprehension of CPU testing and performance analysis. Working with my group members to explore several benchmarking tools such as Geekbench, CPU-Z, Windows Task Manager, and MiniTool Partition Wizard gave me useful knowledge about how computer hardware and software interact. The practical method of comparing various systems and evaluating the information improved my technical skills and enhanced my capacity to communicate complex ideas clearly.

**NABIL AFLAH BOO –** Throughout this project, I have gotten the proper perception of the CPU benchmarking systems. I personal discovered how to test portions of the CPU performance through specific tools such as CPU-Z, Geekbench, Monitor Task Manager, and MiniTool Partition Wizard. Frequently upgrading the CPU performance is important for attaining a higher processor quality and faster performance of the applications. I also use the benchmarking results to coordinate with my teammates on which laptops are the best. I also get a chance to bolster up the technical skills as well as the communication skills as a concept is shared amongst the team members.

**NURUL ADRIANA -** This project has been an extremely beneficial learning experience for me, considerably improving my comprehension of CPU testing and performance analysis. Collaborating with my group members allowed us to explore a variety of testing programmes, including GeekBench, CPU-Z, Windows Task Manager, and MiniTool Partition Wizard. This hands-on approach revealed important insights into the connection of computer hardware and software. Practical comparisons of different systems helped me strengthen my technical talents and communication skills, allowing me to communicate complicated concepts more effectively.

**HARINI A/P SANGARAN –** I have learned how to use the 4 benchmarking tools used in this project which are CPU-Z, Geekbench, Task Manager and Minitool Partition Wizard applications. I think it is vital to know how to use these tools as they allow me to make decisions based on my computer's performance. Apart from this project, I have also learned how to use asm coding which is a new programming language for me. Overall, I think that this project and the subject has widen my knowledge.

- ## Reference

1. Benchmarks as Fast As Possible
   Link : https://youtu.be/cHtNJngtS1I

2. Best CPU Benchmark Software For Windows
   Link : https://youtu.be/fgWGtWsDXmw

3. First 5 Things I Do When Benchmarking.
   Link : https://youtu.be/19eUZ0GwTxI

4. What's the Difference Between a CPU and a GPU?
   https://blogs.nvidia.com/blog/2009/12/16/whats-the-differencebetween-a-cpu-and-a-gpu

5. Zak, R. (2021, February 11). How to use CPU-Z: the complete guide https://www.techradar.com/how-to/how-to-use-cpu-z

6. How to benchmark your GPU on Window 10 https://youtu.be/sLYjwnJO2_w?si=pAMkkBQXxLNb6v1_

## Task Distribution

Member 1- Harini
Flowchart of Project (benchmarking methodology / procedure)
Benchmarking results
Analysis, comparison, discussion
Appendices (Tasks distribution, group pictures)

Member2 - Jingyi
Coding
Benchmarking results
Analysis, comparison, discussion (CPU-Z)

Member3 - Aflah
Introduction
Benchmarking results
Analysis, comparison, discussion (minitool)

Member 4 - Adriana
Conclusion and reflection
Benchmarking results
Analysis, comparison, discussion (geekbench)

- # Appendix

From left: Adriana, Harini, Aflah, Jingyi